Mobile Computing
MANET

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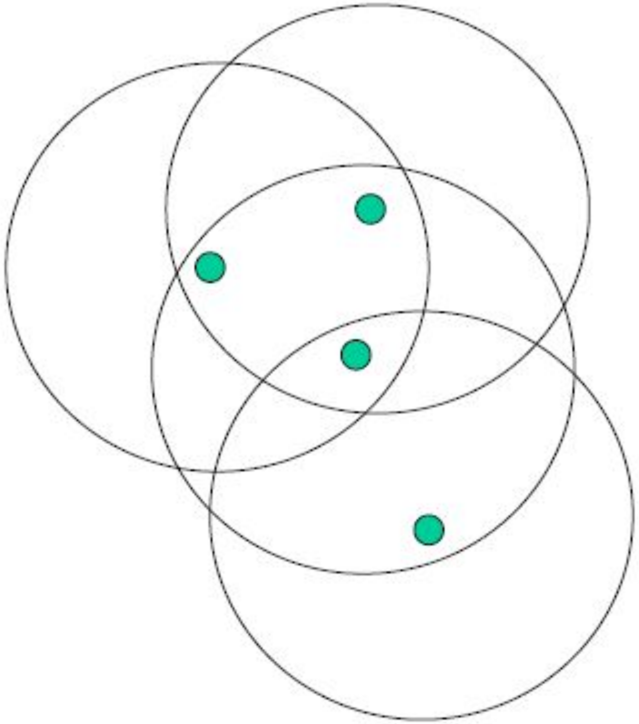
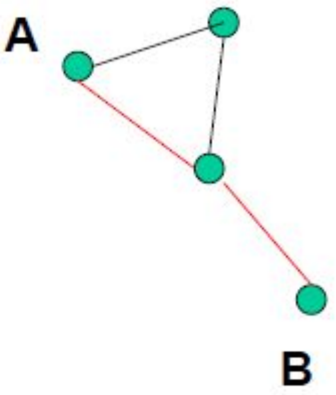
MANET

- Formed by wireless hosts which may be mobile
- Without (necessarily) using a pre-existing infrastructure
- Routes between nodes may potentially contain multiple hops

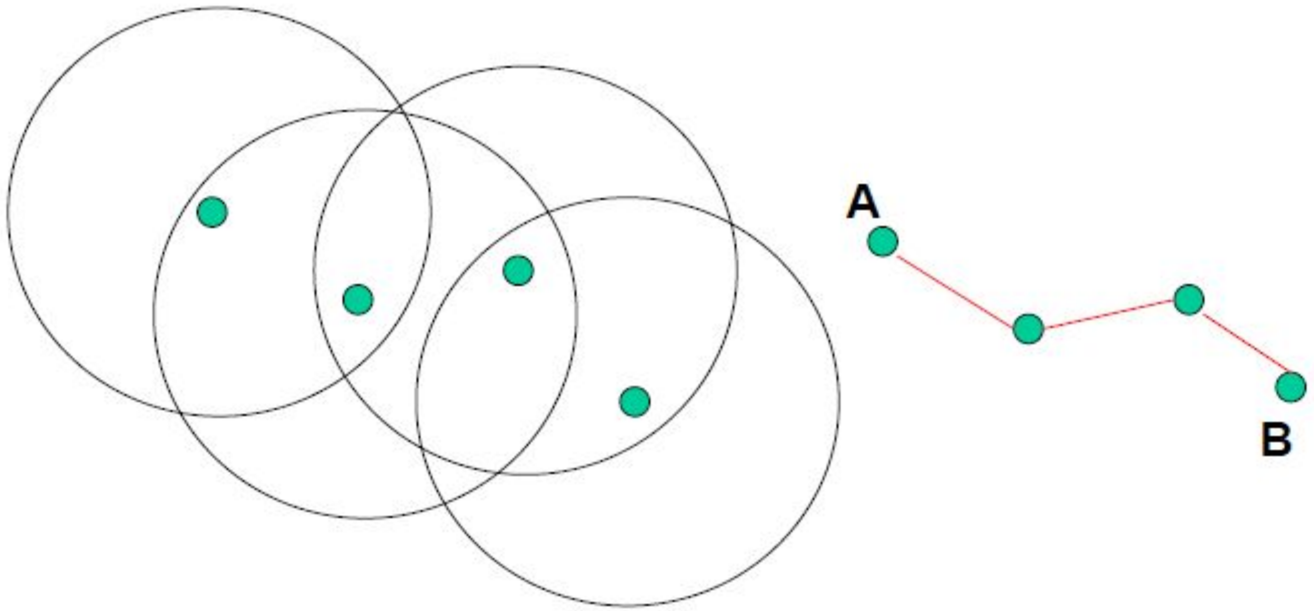
MANET

- May need to traverse multiple links to reach a destination





MANET (Route Changes: Mobility)



Why Ad-hoc Networks

- Ease of deployment
- Speed of deployment
- Decreased dependence on infrastructure

Many Applications

- Personal area networking
  - cell phone, laptop, ear phone, wrist watch
- Military environments
  - soldiers, tanks, planes
- Civilian environments
  - Mesh networks
  - taxi cab network
  - meeting rooms
  - sports stadiums
  - boats, small aircraft
- Emergency operations
  - search-and-rescue
  - policing and fire fighting

Many Variations

- Fully Symmetric Environment
  - all nodes have identical capabilities and responsibilities
- Asymmetric Capabilities
  - transmission ranges and radios may differ
  - battery life at different nodes may differ
  - processing capacity may be different at different nodes
  - speed of movement
- Asymmetric Responsibilities
  - only some nodes may route packets
  - some nodes may act as leaders of nearby nodes (e.g. cluster head)

Many Variations

- Traffic characteristics may differ in different ad hoc networks
  - bit rate
  - timeliness constraints
  - reliability requirements
  - unicast / multicast / geocast
  - host-based addressing / content-based addressing /capability-based addressing
- May co-exist (and co-operate) with an infrastructure based network

Many Variations

- Mobility patterns may be different
  - people sitting at an airport lounge
  - taxi cabs
  - kids playing
  - military movements
  - personal area network
- Mobility characteristics
  - speed
  - Predictability
    - direction of movement
    - pattern of movement
- uniformity (or lack) of mobility characteristics among different nodes

Challenges

- Limited wireless transmission range
- Broadcast nature of the wireless medium
- Hidden terminal problem
- Packet losses due to transmission errors
- Mobility-induced route changes
- Mobility-induced packet losses
- Battery constraints
- Potentially frequent network partitions
- Ease of snooping on wireless transmissions (security hazard)

General Assumption

- Unless stated otherwise, fully symmetric environment is assumed implicitly
  - All nodes have identical capabilities and responsibilities

Why Routing in MANET different ?

- Host mobility
  - link failure/repair due to mobility may have different characteristics than those due to other causes
- Rate of link failure/repair may be high when nodes move fast
- New performance criteria may be used
  - route stability despite mobility
  - energy consumption

Unicast Routing Protocols

- Many protocols have been proposed
- Some have been invented specifically for MANET
- Others are adapted from previously proposed protocols for wired networks
- No single protocol works well in all environments
  - some attempts made to develop adaptive protocols

Routing Protocols

- Proactive protocols
  - Determine routes independent of traffic pattern
  - Traditional link-state and distance-vector routing protocols are proactive
- Reactive protocols
  - Maintain routes only if needed
- Hybrid protocols

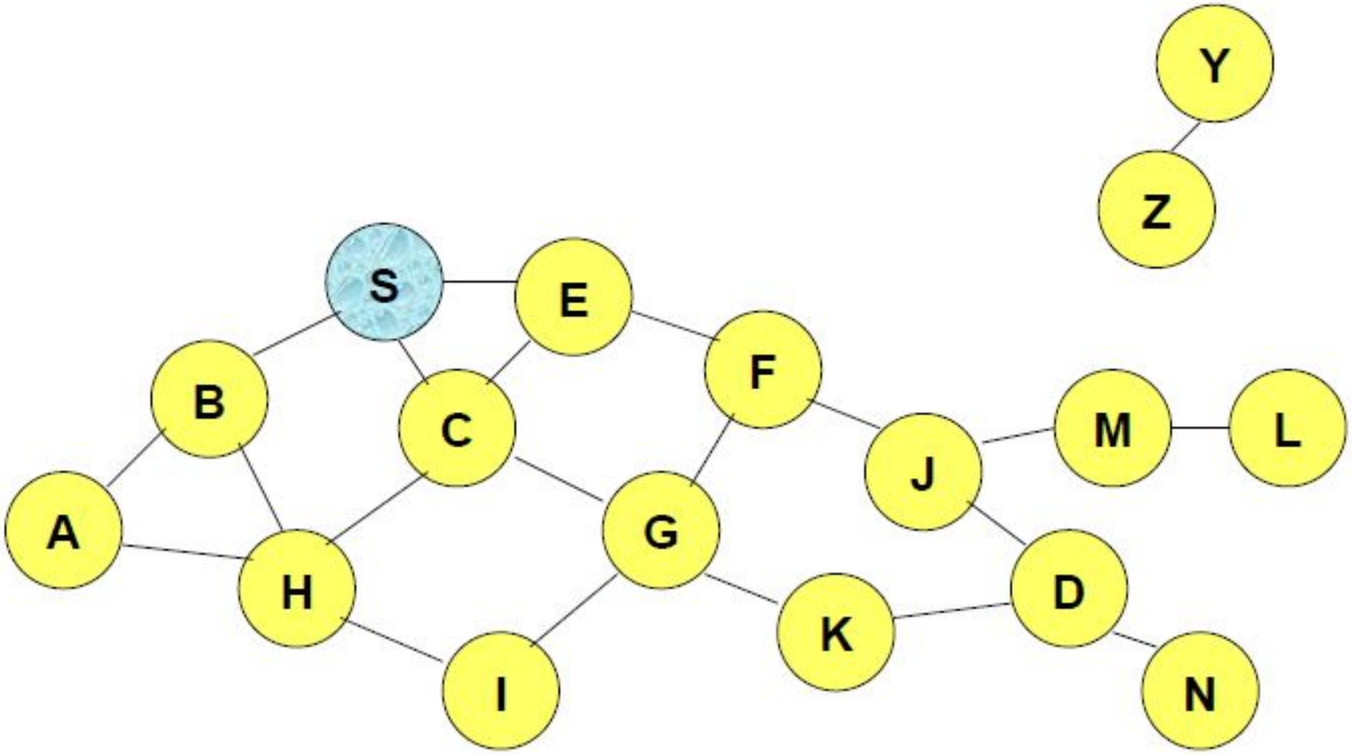
Trade-Off

- Latency of route discovery
  – Proactive protocols may have lower latency since routes are maintained at all times
  – Reactive protocols may have higher latency because a route from X to Y will be found only when X attempts to send to Y
- Overhead of route discovery/maintenance
  – Reactive protocols may have lower overhead since routes are determined only if needed
  – Proactive protocols can (but not necessarily) result in higher overhead due to continuous route updating
- Which approach achieves a better trade-off depends on the traffic and mobility patterns

Flooding

- Sender S broadcasts data packet P to all its neighbors
- Each node receiving P forwards P to its neighbors
- Sequence numbers used to avoid the possibility of forwarding the same packet more than once
- Packet P reaches destination D provided that D is reachable from sender S
- Node D does not forward the packet

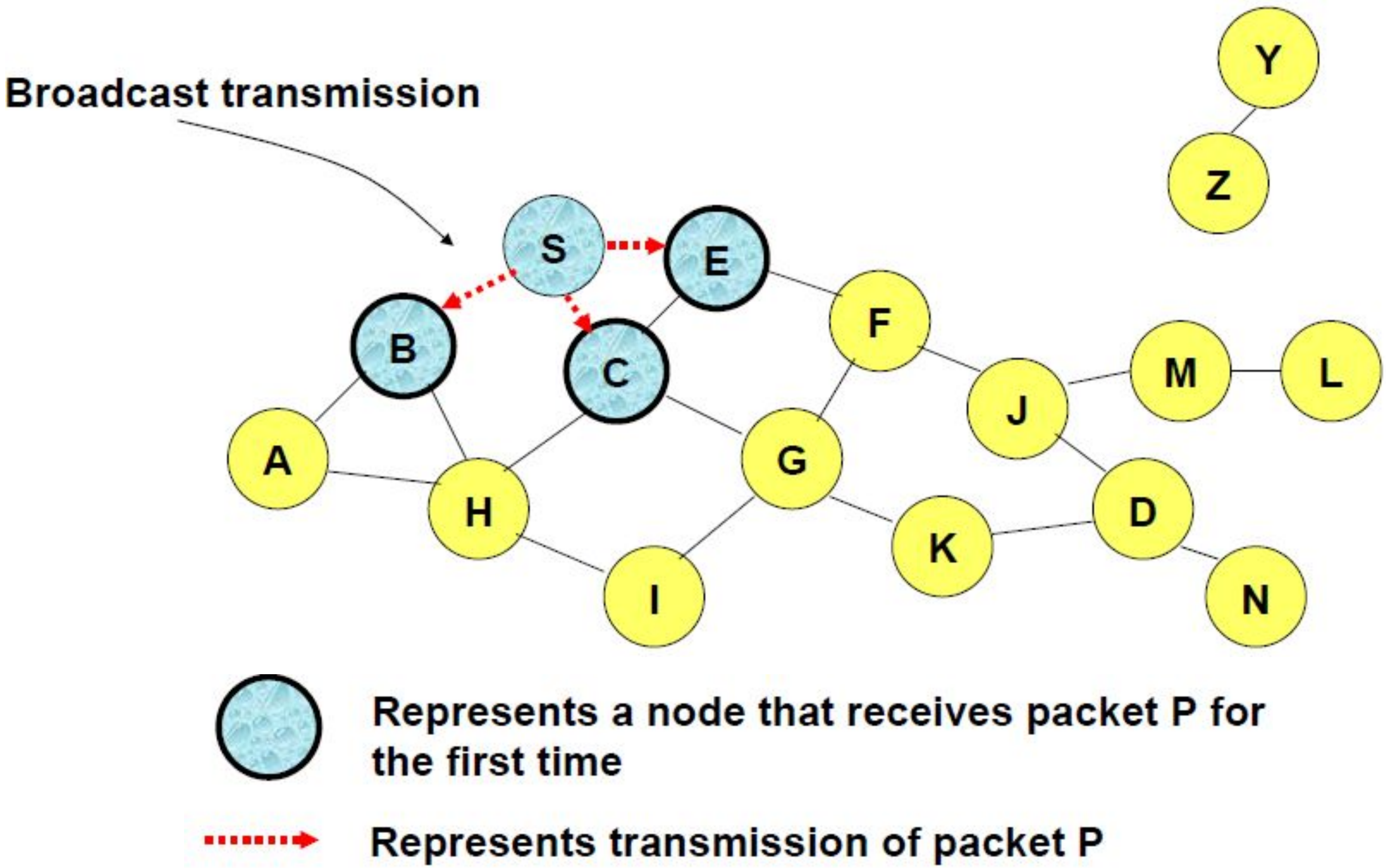
Flooding



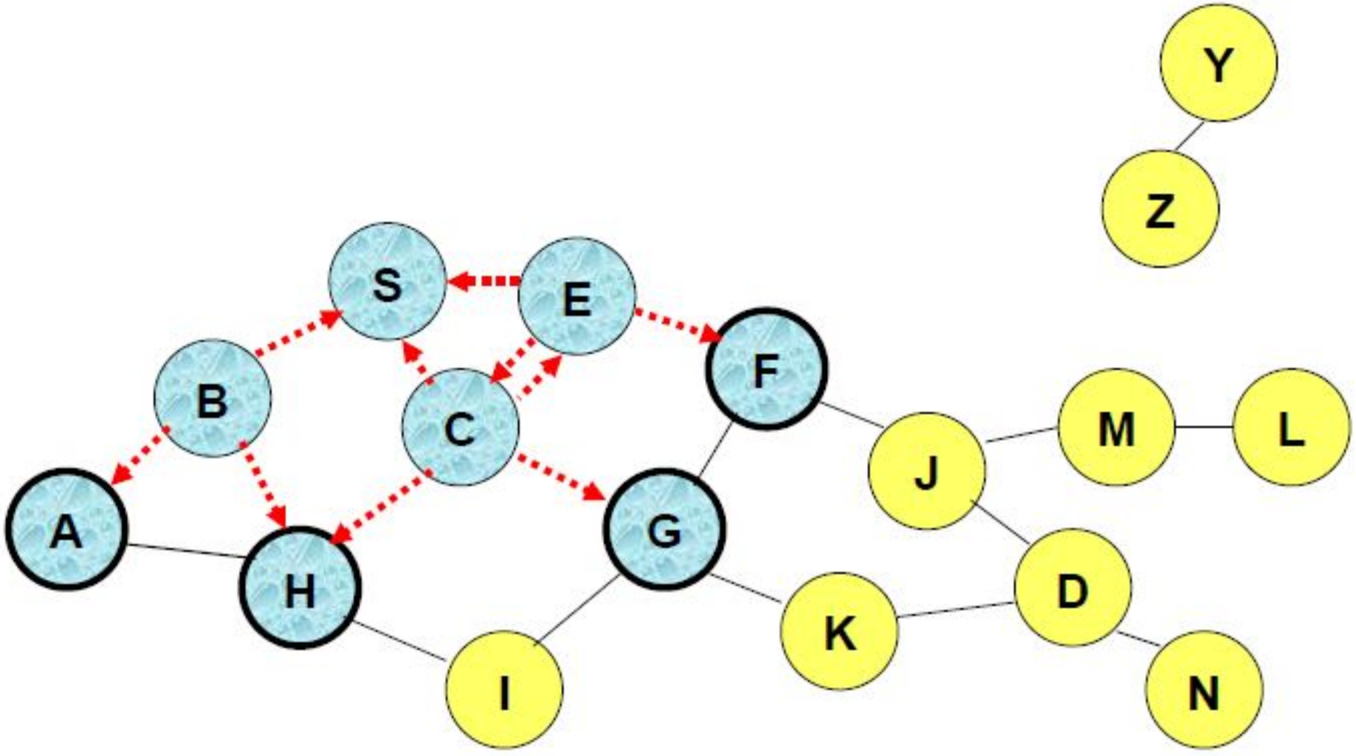
Represents a node that has received packet P

Represents that connected nodes are within each other's transmission range

Flooding

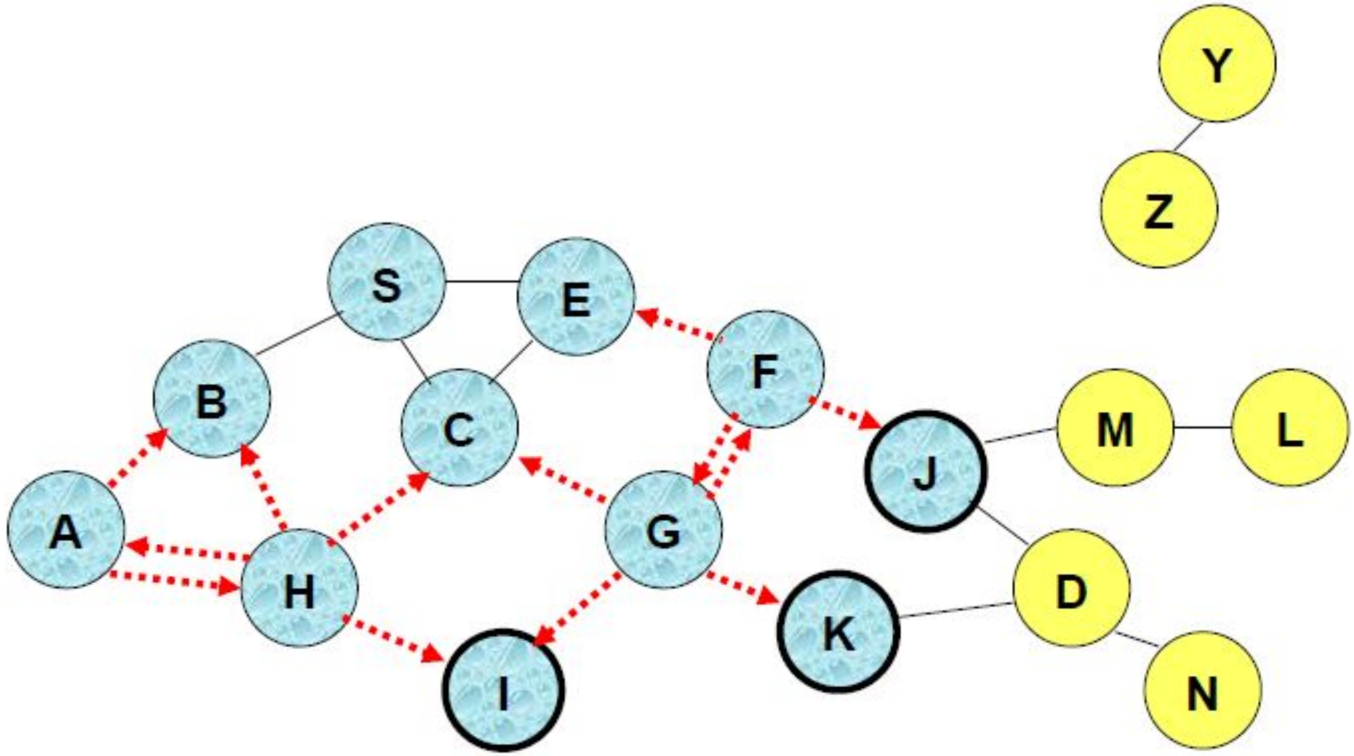

Broadcast transmission
Y
Z
S
E
B
C
F
M
L
J
A
G
H
D
K
I
N
Represents a node that receives packet P for the first time
Represents transmission of packet P

Flooding



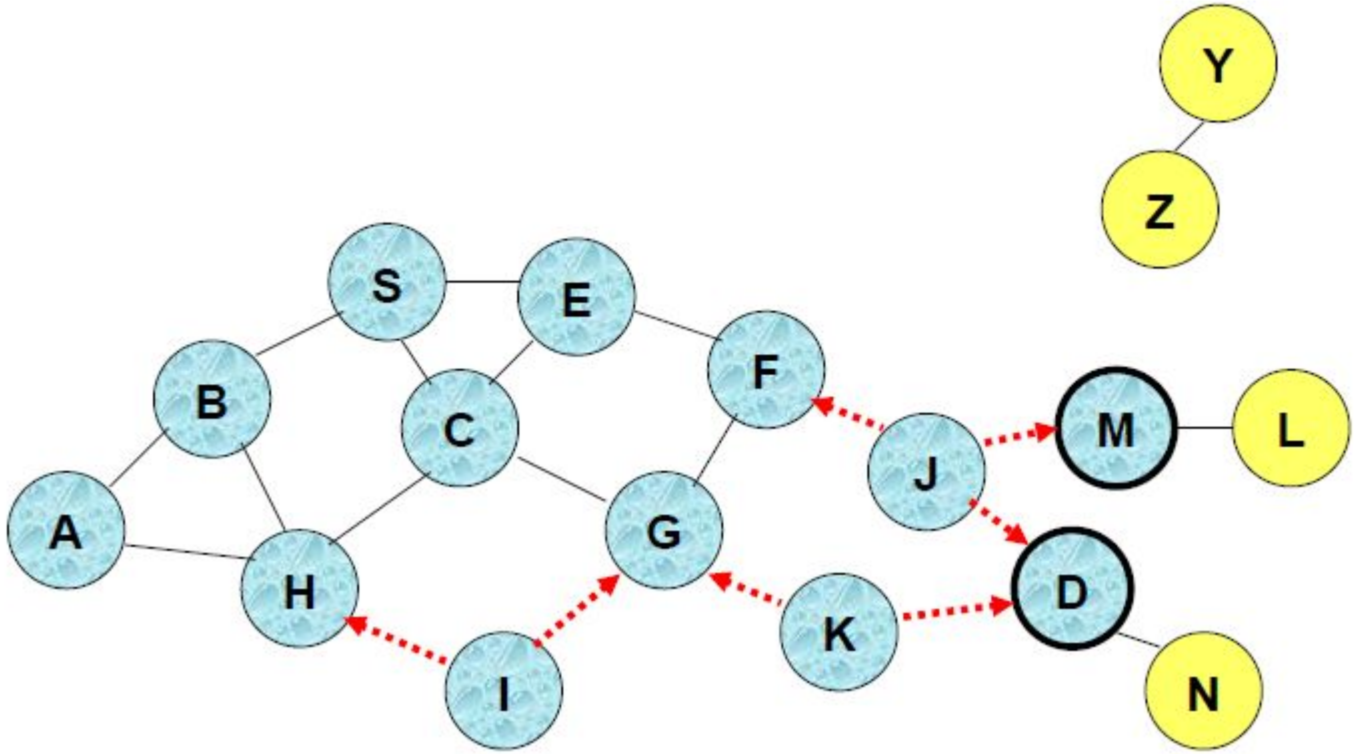
- Node H receives packet P from two neighbors: potential for collision

Flooding



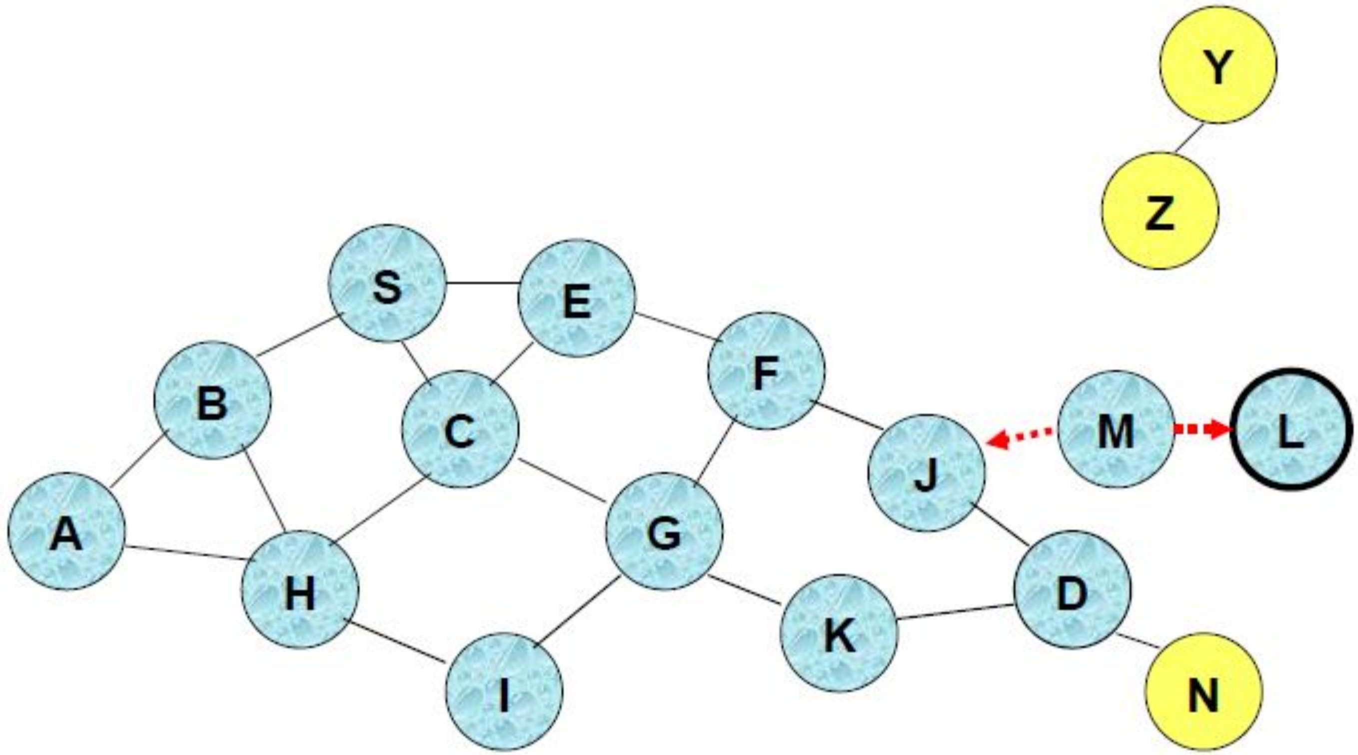
- Node C receives packet P from G and H, but does not forward it again, because node C has already forwarded packet P once

Flooding



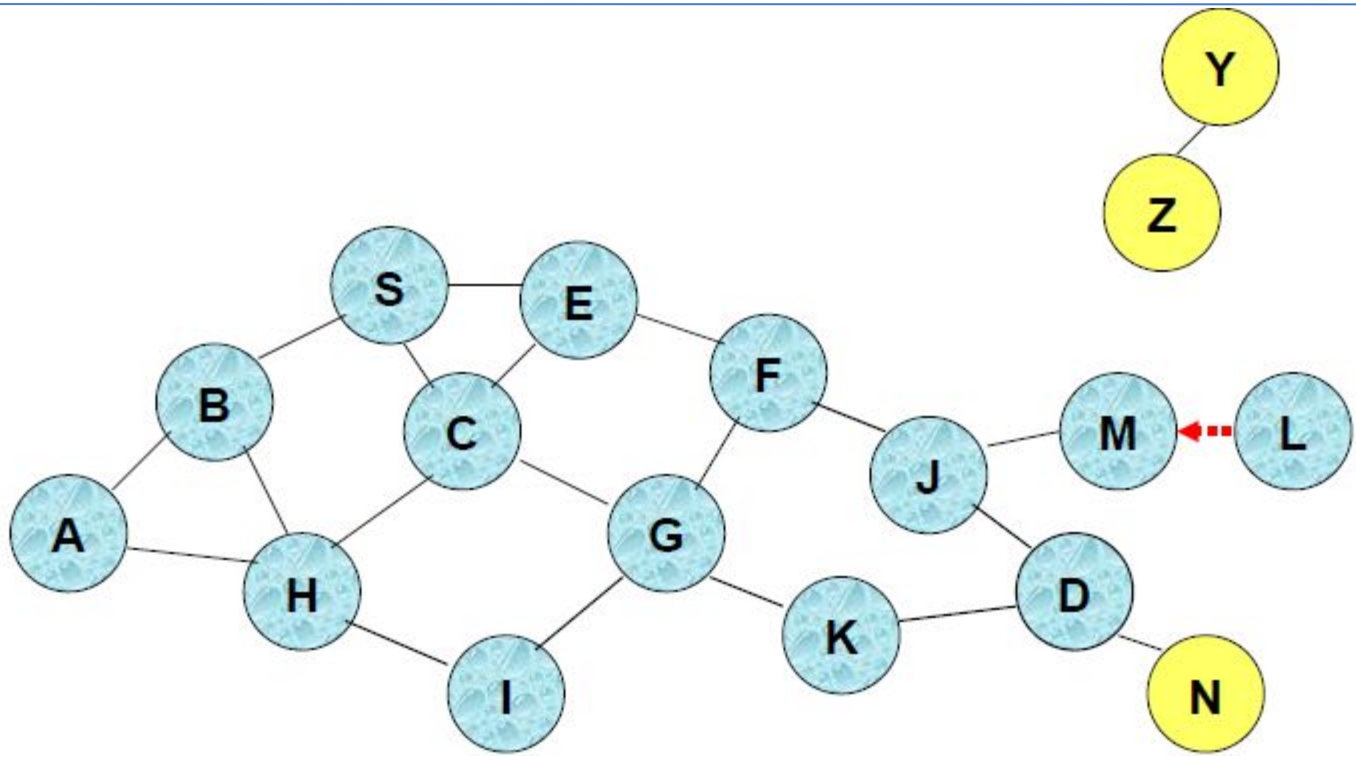
- Nodes J and K both broadcast packet P to node D
  - Since nodes J and K are hidden from each other, their transmissions may collide
    - Packet P may not be delivered to node D at all, despite the use of flooding

Flooding



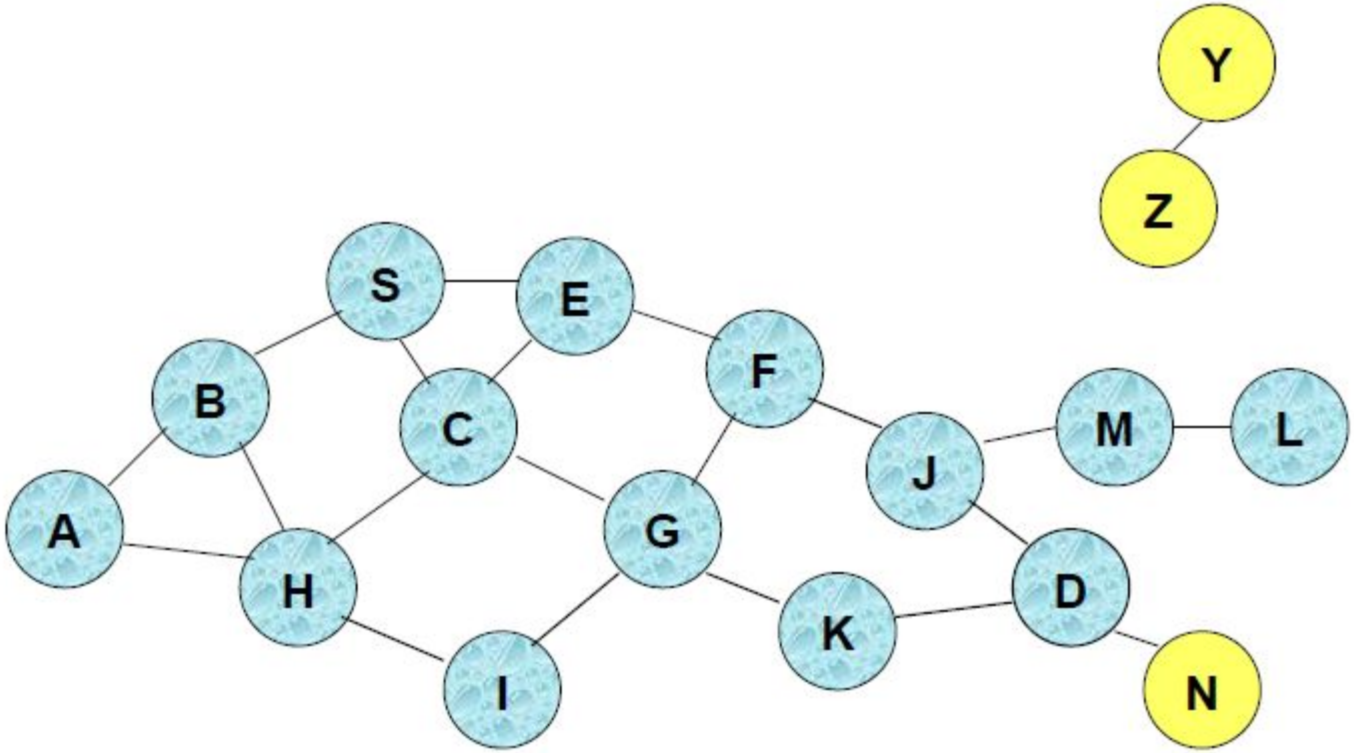
- Node D does not forward packet P, because node D is the intended destination of packet P

Flooding



- Flooding completed
- Nodes unreachable from S do not receive packet P (e.g., node Z)
- Nodes for which all paths from S go through the destination D also do not receive packet P (example: node N)

Flooding



- Flooding may deliver packets to too many nodes (in the worst case, all nodes reachable from sender may receive the packet)

Flooding (Advantages)

- Simplicity
  - May be more efficient than other protocols when rate of information transmission is low enough that the overhead of explicit route discovery/maintenance incurred by other protocols is relatively higher
    - this scenario may occur, for instance, when nodes transmit small data packets relatively infrequently, and many topology changes occur between consecutive packet transmissions
- Potentially higher reliability of data delivery
  - Because packets may be delivered to the destination on multiple paths

Flooding (Disadvantages)

- Potentially, very high overhead
  - Data packets may be delivered to too many nodes who do not need to receive them
- Potentially lower reliability of data delivery
  - Flooding uses broadcasting -- hard to implement reliable broadcast delivery without significantly increasing overhead
    - Broadcasting in IEEE 802.11 MAC is unreliable
- In our example, nodes J and K may transmit to node D simultaneously, resulting in loss of the packet
  - in this case, destination would not receive the packet at all

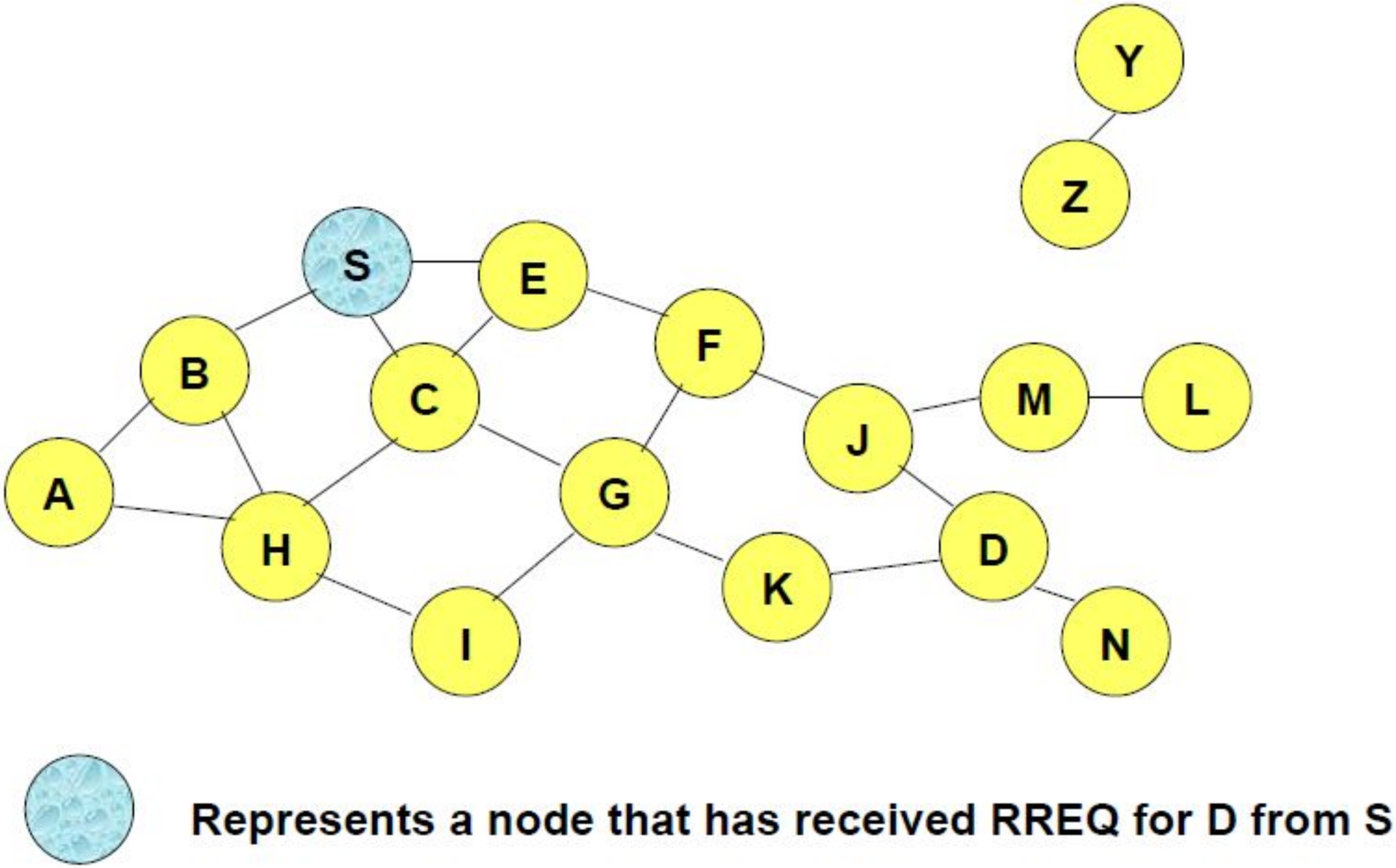
Flooding of Control Packets

- Many protocols perform (potentially *limited) flooding of* control packets, instead of data packets
- The control packets are used to discover routes
- Discovered routes are subsequently used to send data packet(s)
- Overhead of control packet flooding is amortized over data packets transmitted between consecutive control packet floods

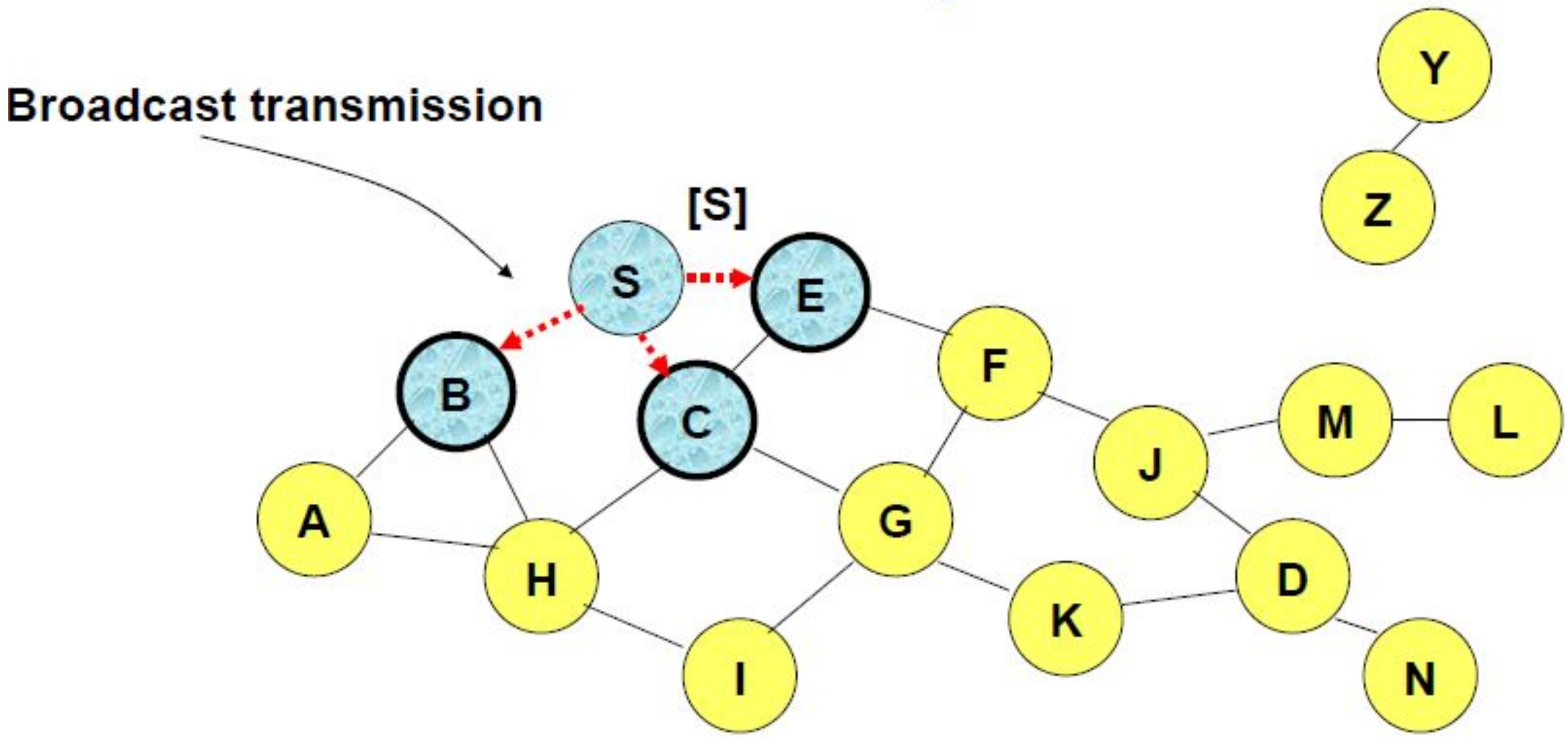
Dynamic Source Routing (DSR)

- When node S wants to send a packet to node D, but does not know a route to D, node S initiates a route discovery
- Source node S floods Route Request (RREQ)
- Each node appends own identifier when forwarding RREQ

Route Discovery (DSR)



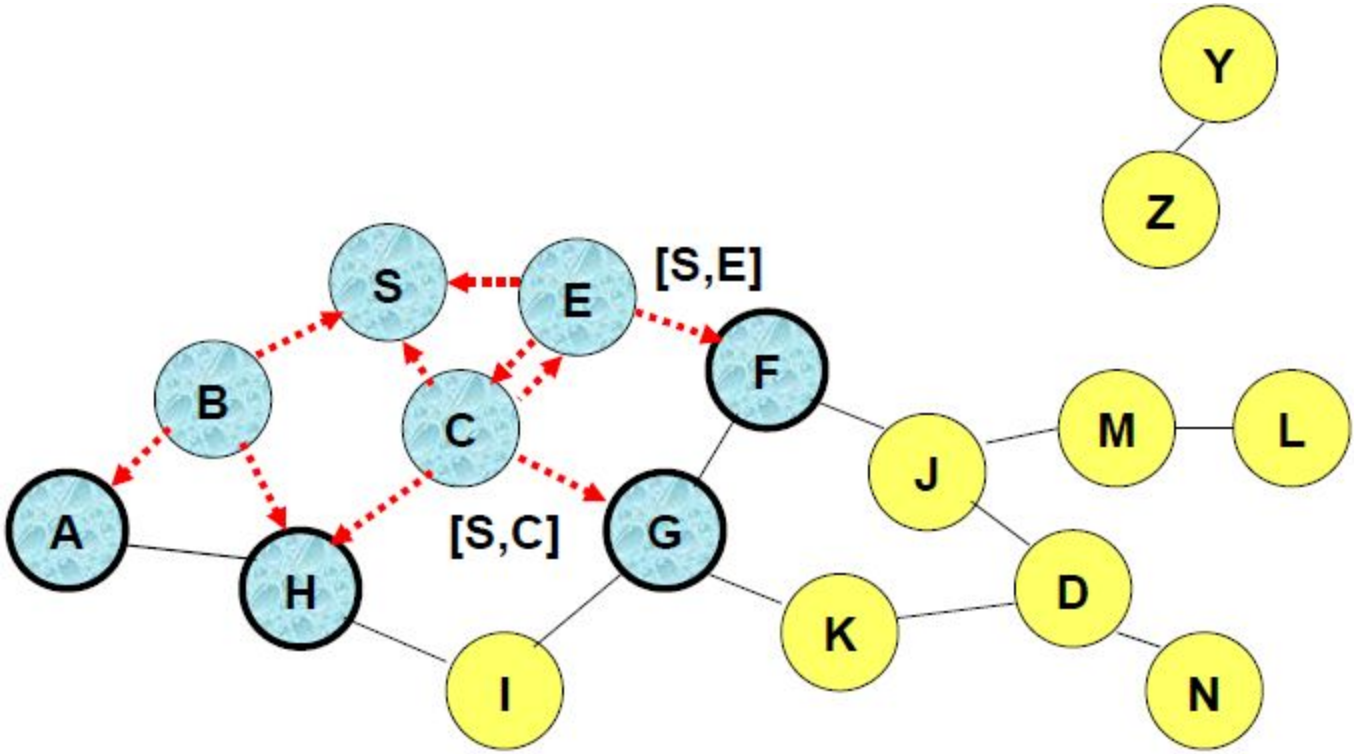
Route Discovery (DSR)



Represents transmission of RREQ

[X,Y] Represents list of identifiers appended to RREQ

Route Discovery (DSR)



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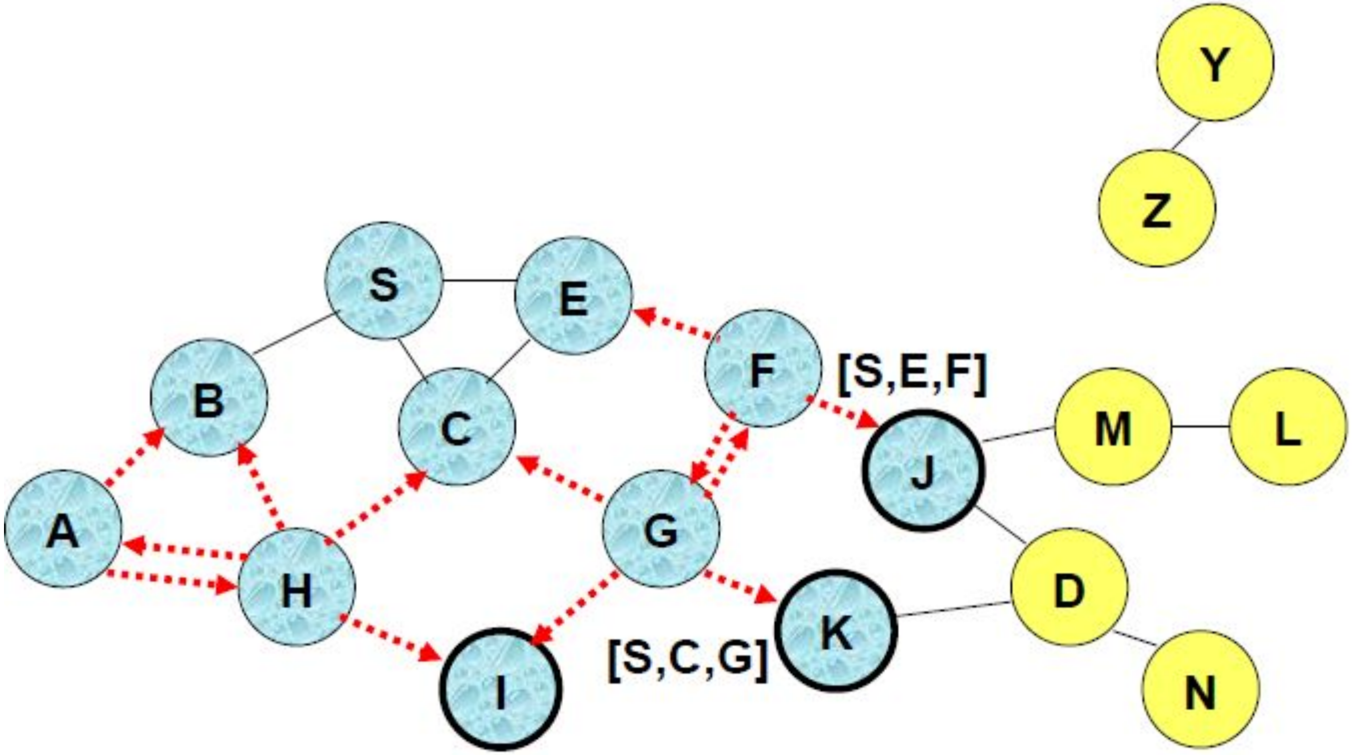
Route Discovery (DSR)



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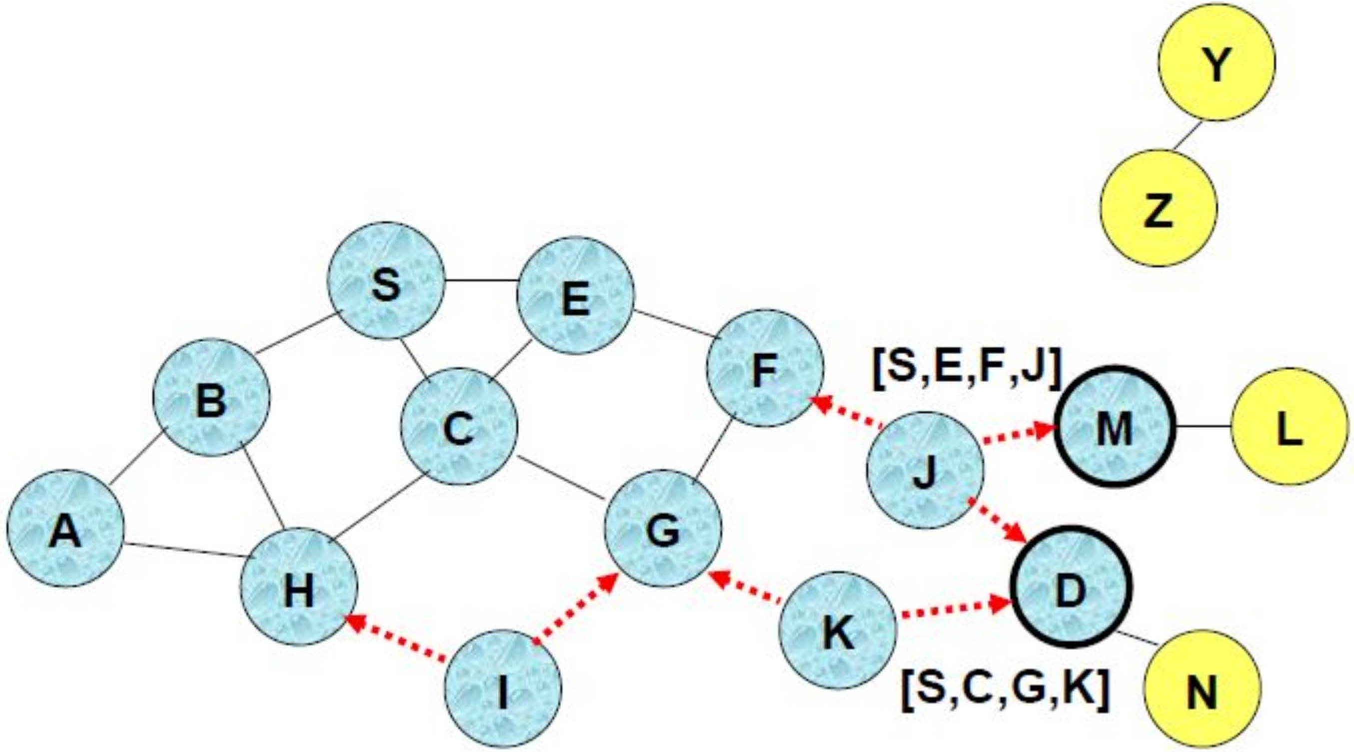
Route Discovery (DSR)



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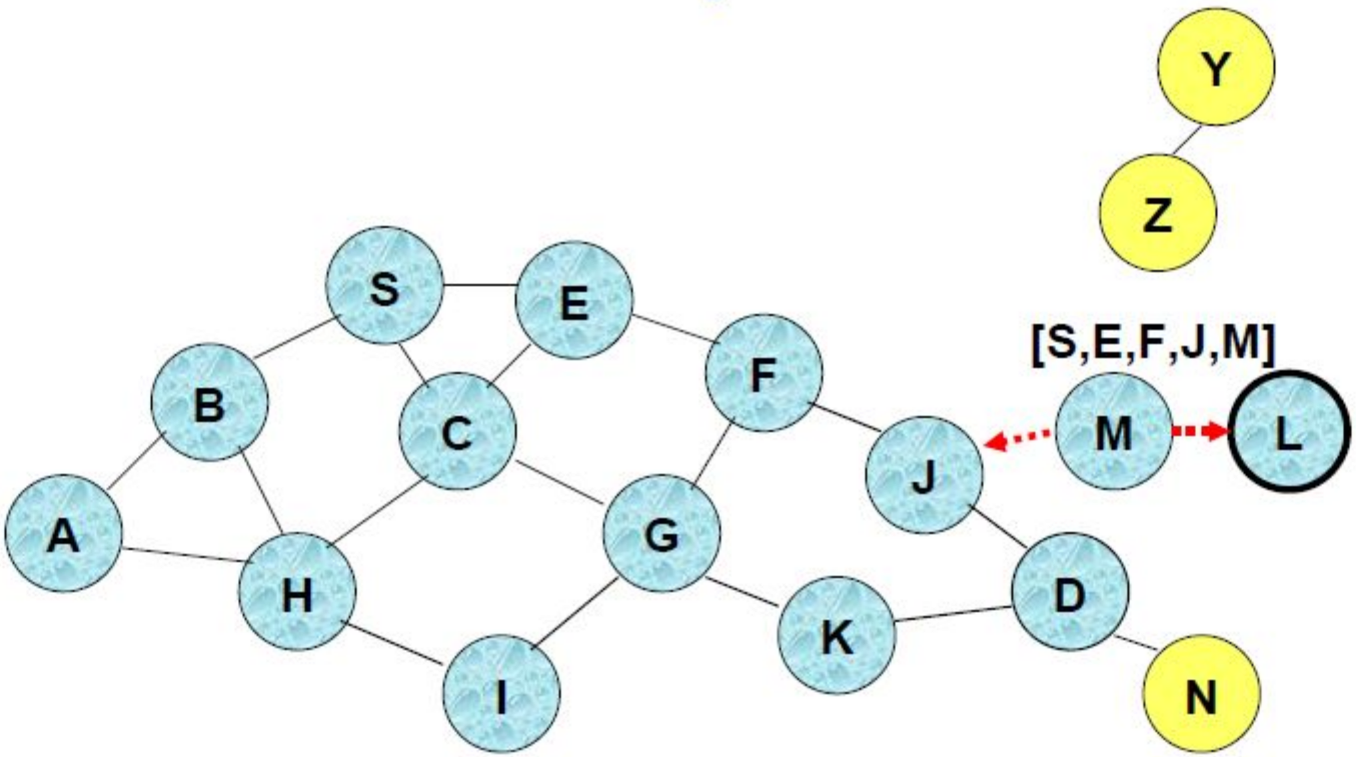
Route Discovery (DSR)

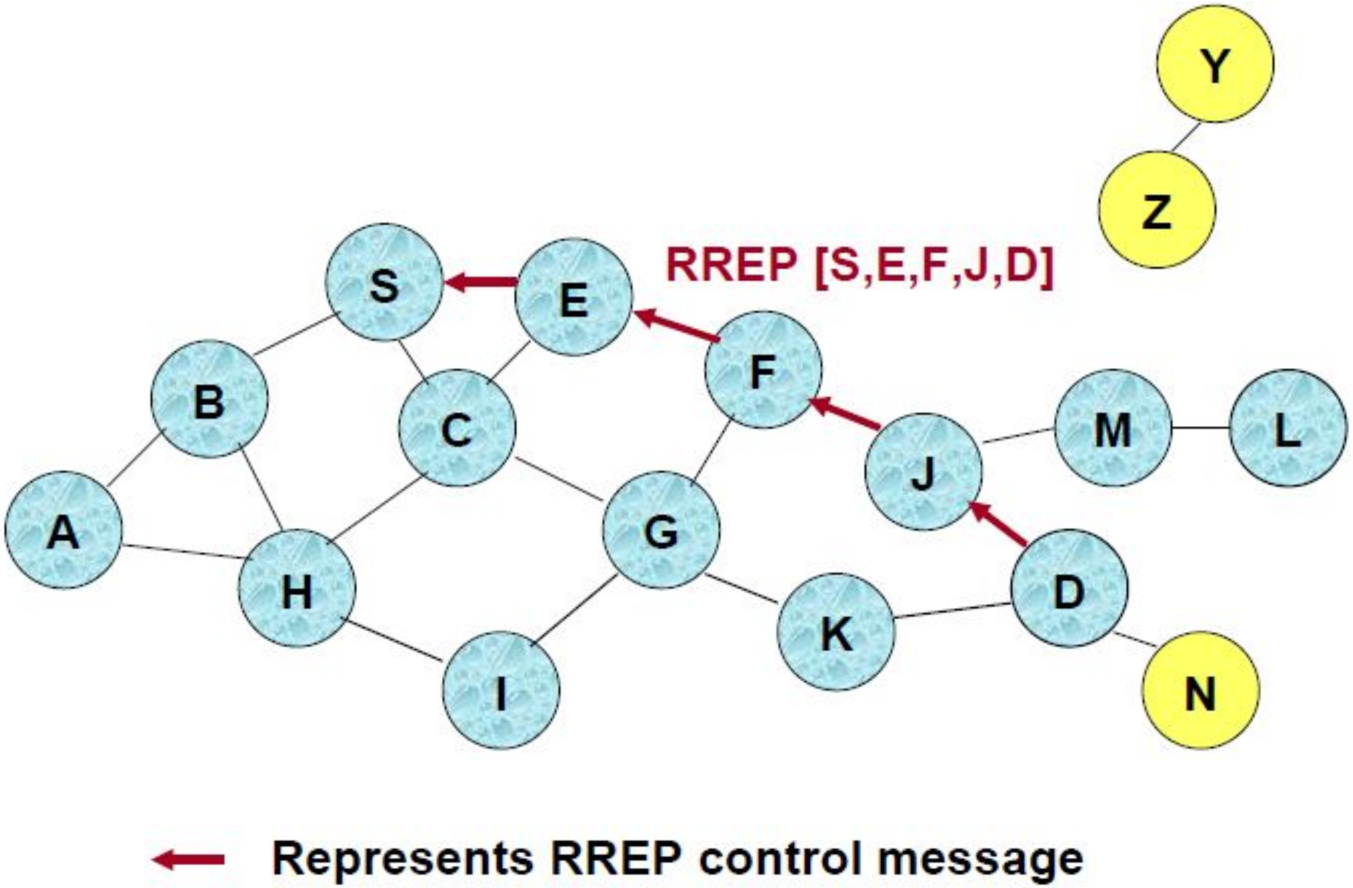


- Node D **does not forward** RREQ, because node D is the **intended target** of the route discovery

Route Discovery (DSR)

- Destination D on receiving the first RREQ, sends a Route Reply (RREP)
- RREP is sent on a route obtained by reversing the route appended to received RREQ
- RREP includes the route from S to D on which RREQ was received by node D

Route Reply (DSR)



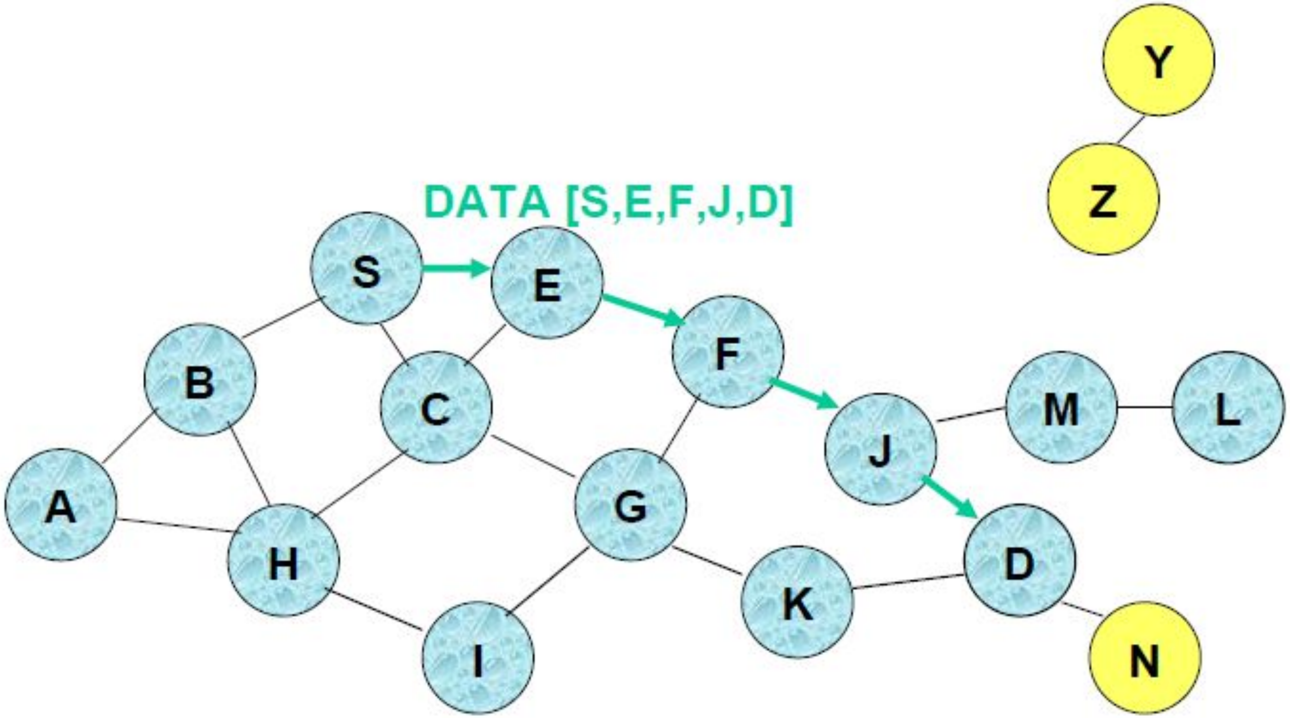
Route Reply (DSR)

- Route Reply can be sent by reversing the route in Route Request (RREQ) only if links are guaranteed to be bi-directional
  - To ensure this, RREQ should be forwarded only if it received on a link that is known to be bi-directional
- If unidirectional (asymmetric) links are allowed, then RREP may need a route discovery for S from node D
  - Unless node D already knows a route to node S
  - If a route discovery is initiated by D for a route to S, then the Route Reply is piggybacked on the Route Request from D.
- If IEEE 802.11 MAC is used to send data, then links have to be bi-directional (since Ack is used)

Dynamic Source Routing (DSR)

- Node S on receiving RREP, caches the route included in the RREP
- When node S sends a data packet to D, the entire route is included in the packet header
  - hence the name source routing
- Intermediate nodes use the source route included in a packet to determine to whom a packet should be forwarded

Data Delivery (DSR)



- Packet header size grows with route length
- When to Perform a Route Discovery
  - When node S wants to send data to node D, but does not know a valid route node D

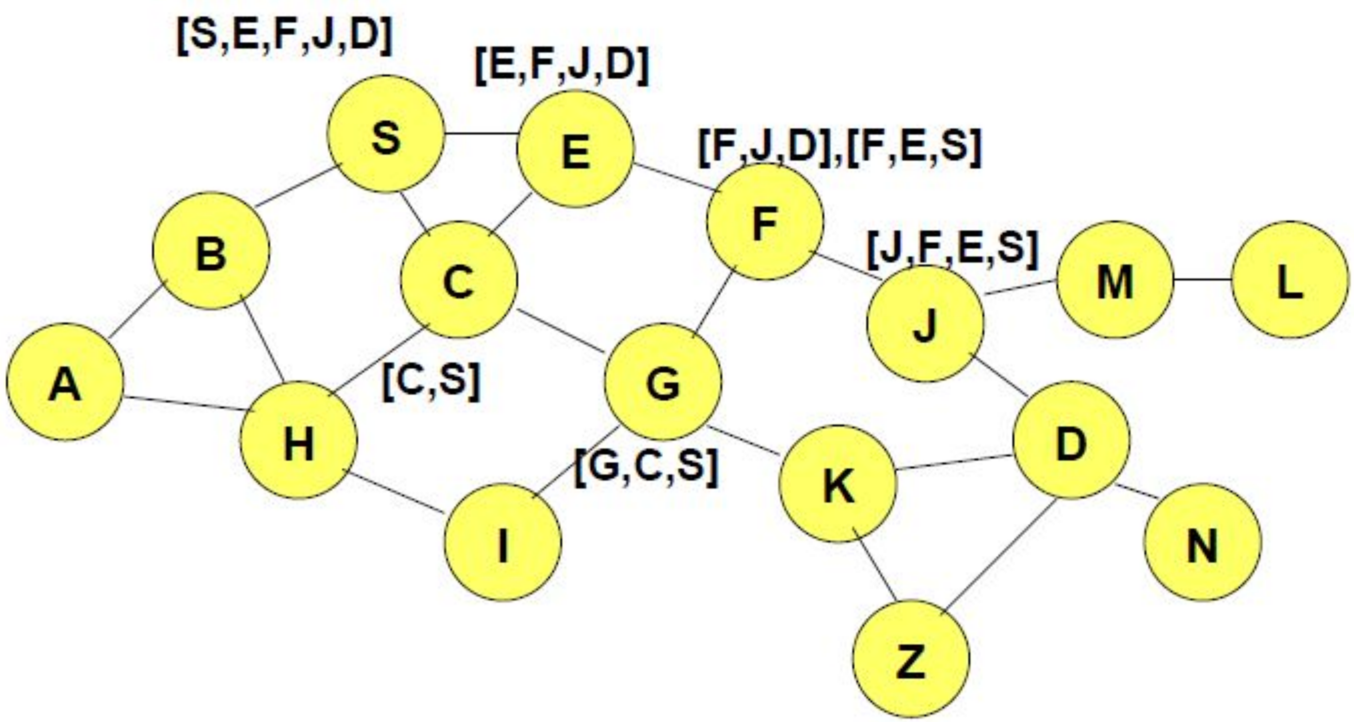
DSR Optimization (Route Caching)

- Each node caches a new route it learns by *any means*
- When node S finds route [S,E,F,J,D] to node D,
  - node S also learns route [S,E,F] to node F
- When node K receives Route Request [S,C,G] destined for node,
  - node K learns route [K,G,C,S] to node S
- When node F forwards Route Reply RREP [S,E,F,J,D],
  - node F learns route [F,J,D] to node D
- When node E forwards Data [S,E,F,J,D]
  - it learns route [E,F,J,D] to node D
- A node may also learn a route when it overhears Data packets

Use of Route Caching

- When node S learns that a route to node D is broken, it uses another route from its local cache, if such a route to D exists in its cache. Otherwise, node S initiates route discovery by sending a route request
- Node X on receiving a Route Request for some node D can send a Route Reply if node X knows a route to node D
- Use of route cache
  - can speed up route discovery
  - can reduce propagation of route requests

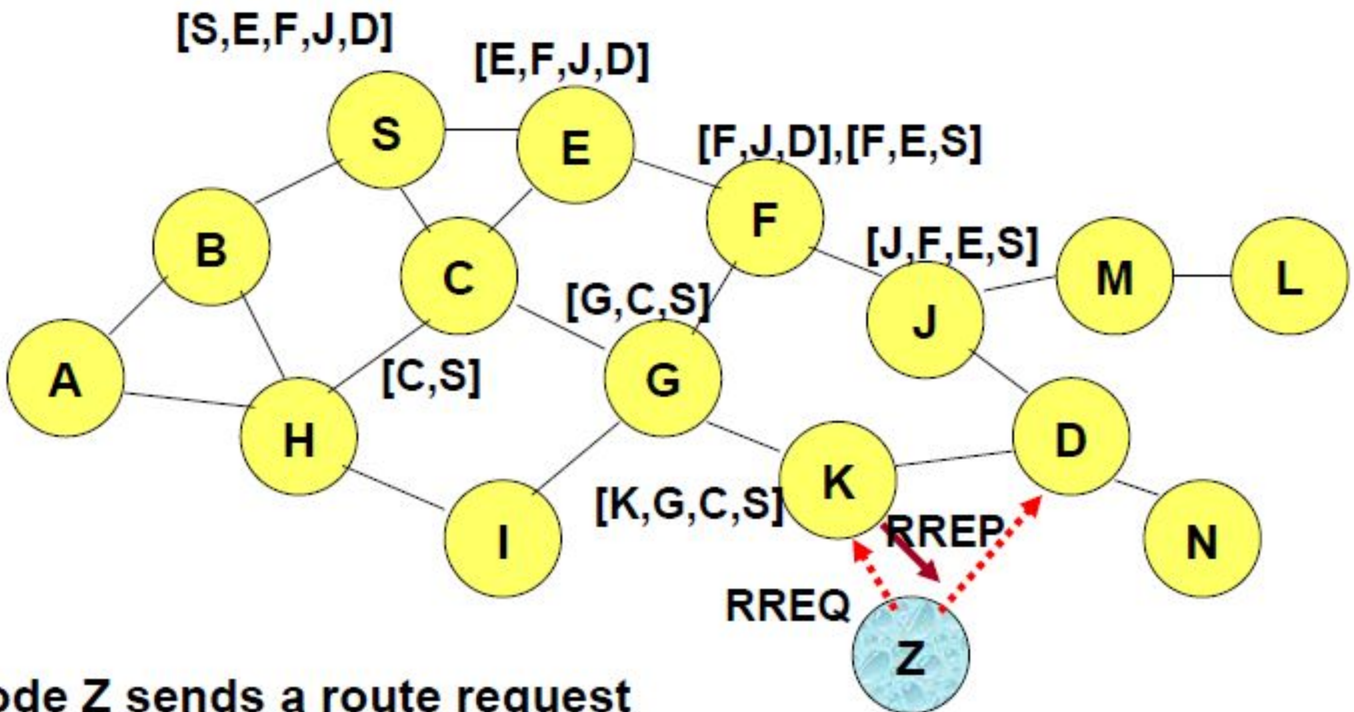
Use of Route Caching



[P,Q,R] Represents cached route at a node
(DSR maintains the cached routes in a tree format)

Use of Route Caching

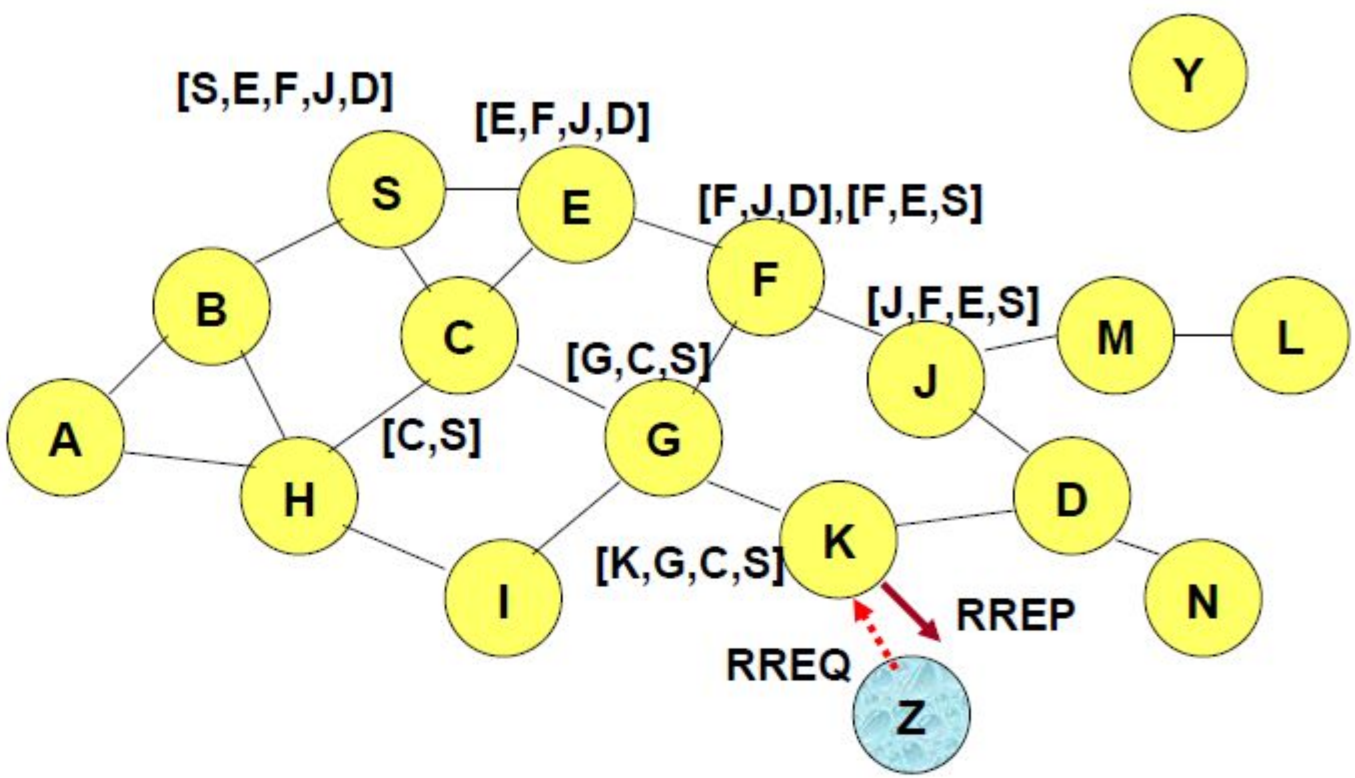
Can Speed up Route Discovery



When node Z sends a route request for node C, node K sends back a route reply [Z,K,G,C] to node Z using a locally cached route

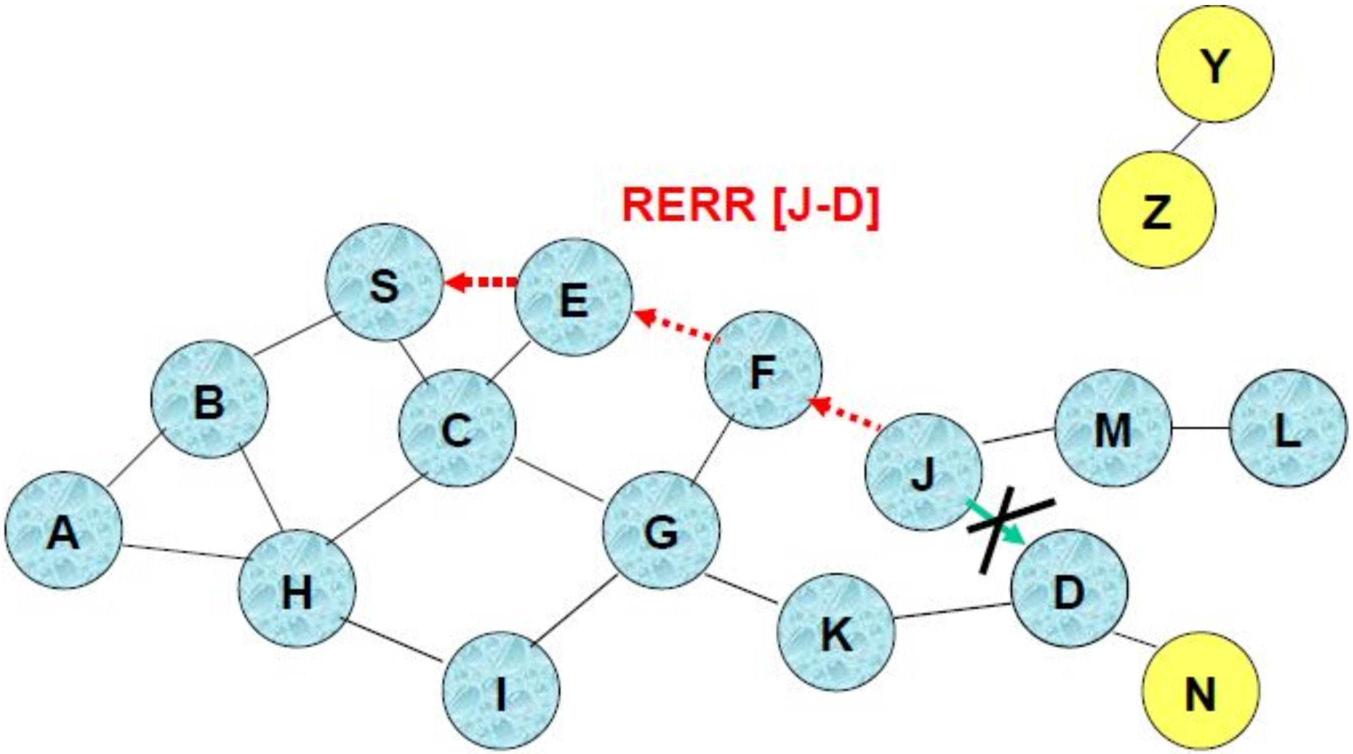
Use of Route Caching

Can Reduce Propagation of Route Requests



Assume that there is no link between D and Z.
Route Reply (RREP) from node K limits flooding of RREQ.
In general, the reduction may be less dramatic.

Route Error (RERR)



J sends a route error to S along route J-F-E-S when its attempt to forward the data packet S (with route SEFJD) on J-D fails

Nodes hearing RERR update their route cache to remove link J-D

Route caching: Cautions

- Stale caches can adversely affect performance
- With passage of time and host mobility, cached routes may become invalid
- A sender host may try several stale routes (obtained from local cache, or replied from cache by other nodes), before finding a good route

DSR (Advantages)

- Routes maintained only between nodes who need to communicate
  - reduces overhead of route maintenance
- Route caching can further reduce route discovery overhead
- A single route discovery may yield many routes to the destination, due to intermediate nodes replying from local caches

DSR (Disadvantages)

- Packet header size grows with route length due to source routing
- Flood of route requests may potentially reach all nodes in the network
- Care must be taken to avoid collisions between route requests propagated by neighboring nodes
  - insertion of random delays before forwarding RREQ

DSR (Disadvantages)

- Increased contention if too many route replies come back due to nodes replying using their local cache
  - Route Reply *Storm problem*
  - Reply storm may be eased by preventing a node from sending RREP if it hears another RREP with a shorter route
- An intermediate node may send Route Reply using a stale cached route, thus polluting other caches
  - This problem can be eased if some mechanism to purge (potentially) invalid cached routes is incorporated.
    - Static timeouts
    - Adaptive timeouts based on link stability

Ad Hoc On-Demand Distance Vector Routing (AODV)

- DSR includes source routes in packet headers
  - Resulting large headers can sometimes degrade performance
    - particularly when data contents of a packet are small
- AODV attempts to improve on DSR by maintaining routing tables at the nodes, so that data packets do not have to contain routes
- AODV retains the desirable feature of DSR that routes are maintained only between nodes which need to communicate

AODV

- Route Requests (RREQ) are forwarded in a manner similar to DSR
- When a node re-broadcasts a Route Request, it sets up a reverse path pointing towards the source
  - AODV assumes symmetric (bi-directional) links
- When the intended destination receives a Route Request, it replies by sending a Route Reply
- Route Reply travels along the reverse path set-up when Route Request is forwarded

AODV (Properties)

- Route discovery cycle used for route finding
  - Maintenance of active routes
  - Sequence numbers used for loop prevention and as route freshness criteria
- Whenever routes are not used, get expired they are discarded
  - Reduces stale routes
  - Reduces need for route maintenance
  - Minimizes number of active routes between an active source and destination

AODV (Properties)

- AODV discovers routes as and when necessary
  - Does not maintain routes from every node to every other
  - Routes are maintained just as long as necessary
- Every node maintains its monotonically increasing sequence number -> increases every time the node notices change in the neighborhood topology

AODV (Properties)

- AODV utilizes routing tables to store routing information
  1. A Routing table for unicast routes
  2. A Routing table for multicast routes

- The route table stores: <destination addr, next-hop addr, destination sequence number, life_time>

- For each destination, a node maintains a list of precursor nodes, to route through them
  - Precusor nodes help in route maintenance (more later)

- Life-time updated every time the route is used
  - If route not used within its life time -> it expires

AODV (Route Discovery)

- When a node wishes to send a packet to some destination -
  - It checks its routing table to determine if it has a current route to the destination
    - If Yes, forwards the packet to next hop node
    - If No, it initiates a route discovery process
- Route discovery process begins with the creation of a Route Request (RREQ) packet -> source node creates it
- The packet contains - source node's IP address, source node's current sequence number, destination IP address, destination sequence number

AODV (Route Discovery)

- Packet also contains broadcast ID number
  - Broadcast ID gets incremented each time a source node uses RREQ
  - Broadcast ID and source IP address form a unique identifier for the RREQ

- Broadcasting is done via Flooding

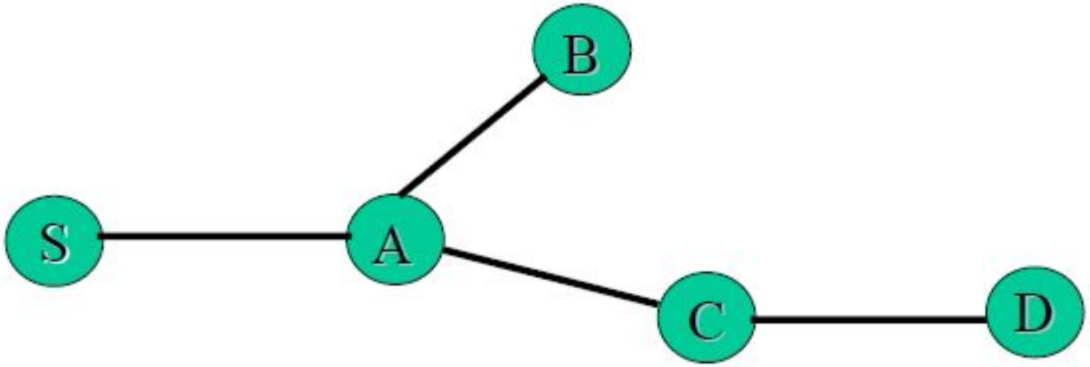
AODV (Route Discovery)

- Once an intermediate node receives a RREQ, the node sets up a reverse route entry for the source node in its route table
  - Reverse route entry consists of <Source IP address, Source seq. number, number of hops to source node, IP address of node from which RREQ was received>
  - Using the reverse route a node can send a RREP (Route Reply packet) to the source
    - Reverse route entry also contains - life time field
- RREQ reaches destination -> In order to respond to RREQ a node should have in its route table:
  1. Unexpired entry for the destination
  2. Seq. number of destination at least as great as in RREQ (for loop prevention)

AODV (Route Discovery)

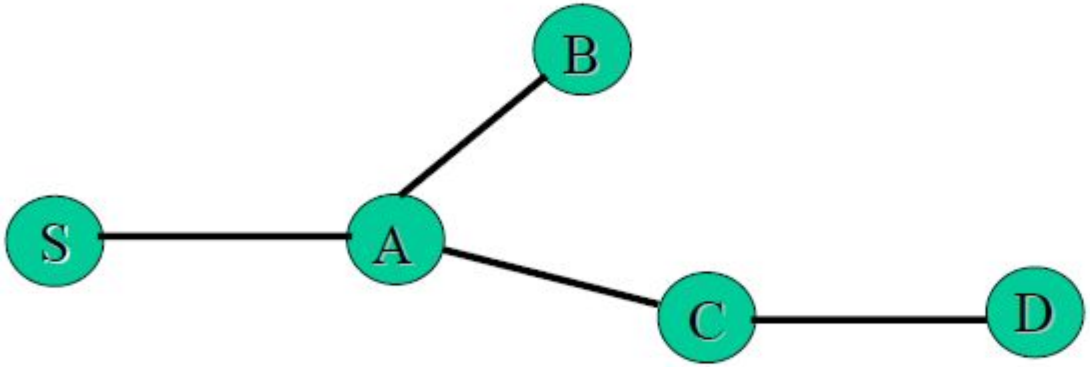
- RREQ reaches destination (contd.)
  - If both conditions are met & the IP address of the destination matches with that in RREQ → the node responds to RREQ by sending a RREP back using unicasting and not flooding to the source using reverse path
  - If conditions are not satisfied, then node increments the hop count in RREQ and broadcasts to its neighbors
- Ultimately the RREQ will make to the destination

Example



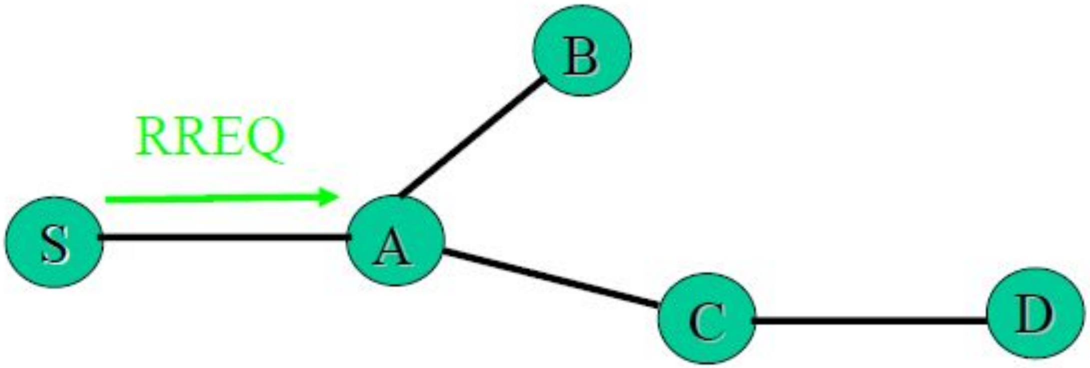
1. Node S needs a route to D

Example



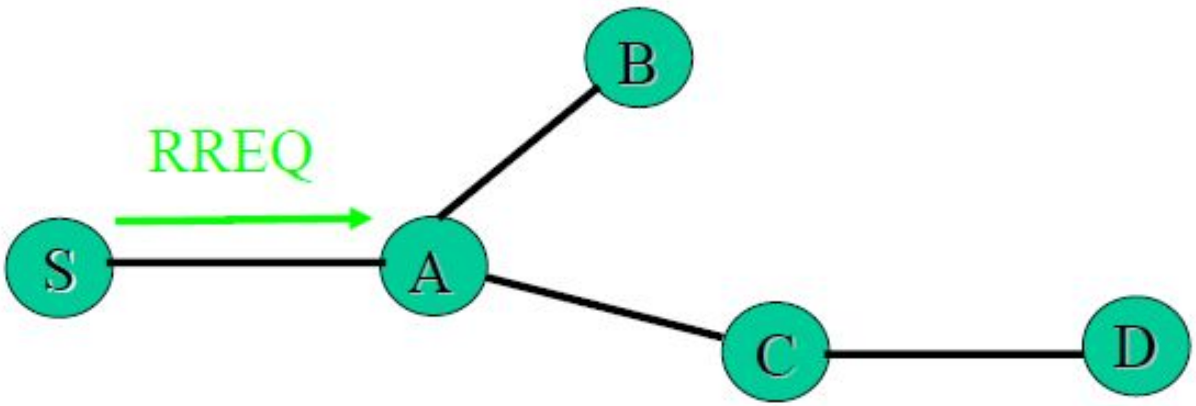
1. Node S needs a route to D
2. Creates a Route Request (RREQ)
   - Enters D's IP addr, seq#, S's IP addr, seq#, hopcount (=0)

Example



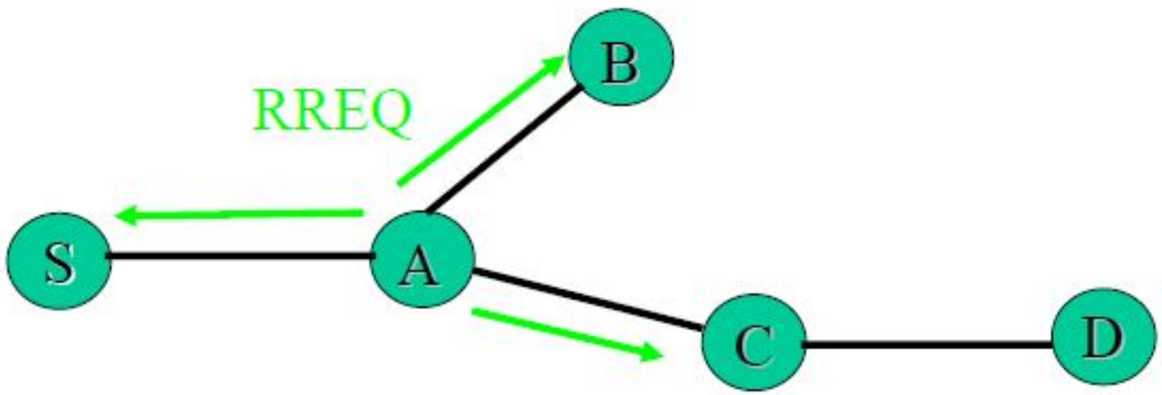
1. Node S needs a route to D
2. Creates a Route Request (RREQ)
   - Enters D's IP addr, seq#, S's IP addr, seq#, hopcount (=0)
3. Node S broadcasts RREQ to neighbors

Example



4. Node A receives RREQ
   - Makes a reverse route entery for S
     dest=S, nexthop=S, hopcount=1
   - It has no routes to D, so it rebroadcasts RREQ

Example



5. Node C receives RREQ
- Makes a reverse route entry for S
  dest=S, nexthop=A, hopcount=2
- It has a route to D, and the seq# for route to D is >= D's seq# in RREQ

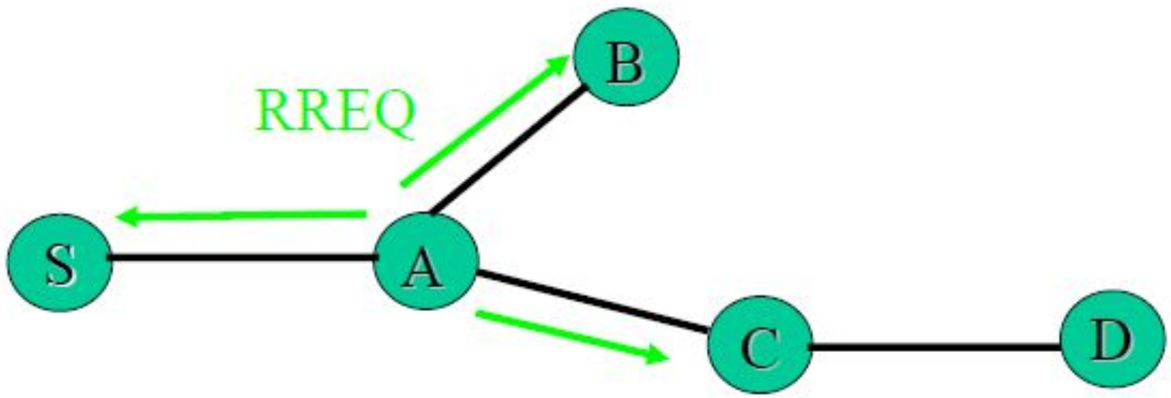
Route Reply AODV (Comments)

- An intermediate node (not the destination) may also send a Route Reply (RREP) provided that it knows a more recent path than the one previously known to sender S

- To determine whether the path known to an intermediate node is more recent, *destination sequence numbers* are used

- The likelihood that an intermediate node will send a Route Reply when using AODV not as high as DSR (Later)
  - A new Route Request by node S for a destination is assigned a higher destination sequence number. An intermediate node which knows a route, but with a smaller sequence number, cannot send Route Reply

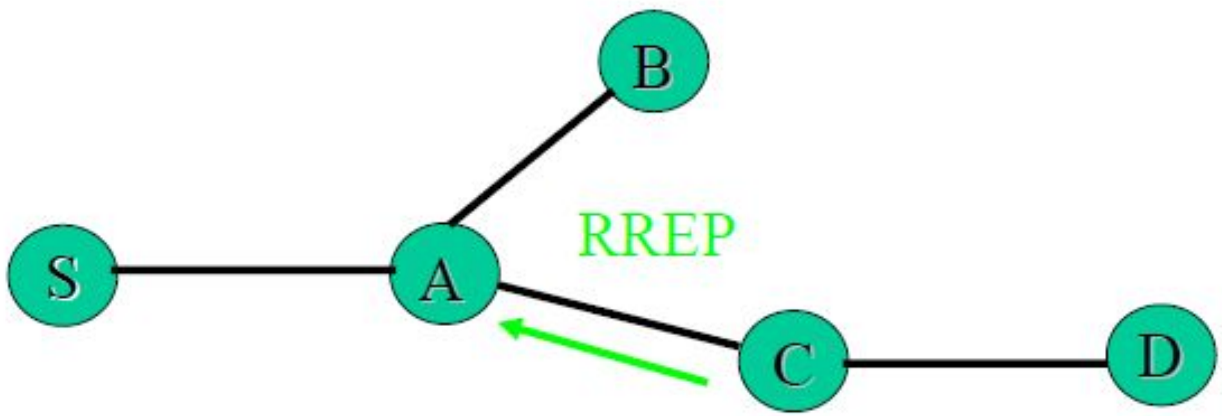
Route Discovery Contd.



5. Node *C* receives RREQ (Cont.)
   - *C* creates a Route Reply (RREP)

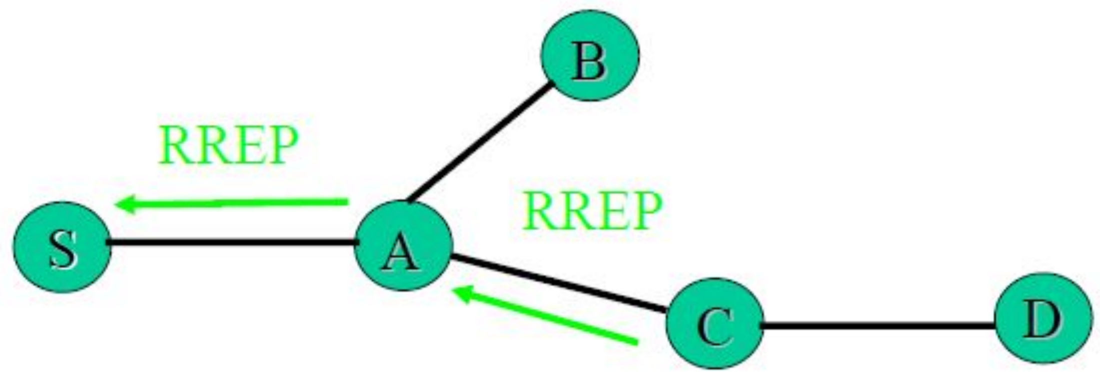
     Enters D's IP addr, seq#, S's IP addr, hopcount to D (=1)
   - Unicasts RREP to *A*

Route Discovery Contd.



5. Node C receives RREQ (Cont.)
   - C creates a Route Reply (RREP)
     Enters D's IP addr, seq#, S's IP addr, hopcount to D (=1)
   - Unicasts RREP to A

Route Discovery Contd.



5. Node A receives RREP
   - Makes a forward route entry to D
     dest=D, nexthop=C, hopcount=2
   - Unicasts RREP to S

Forward Path Setup

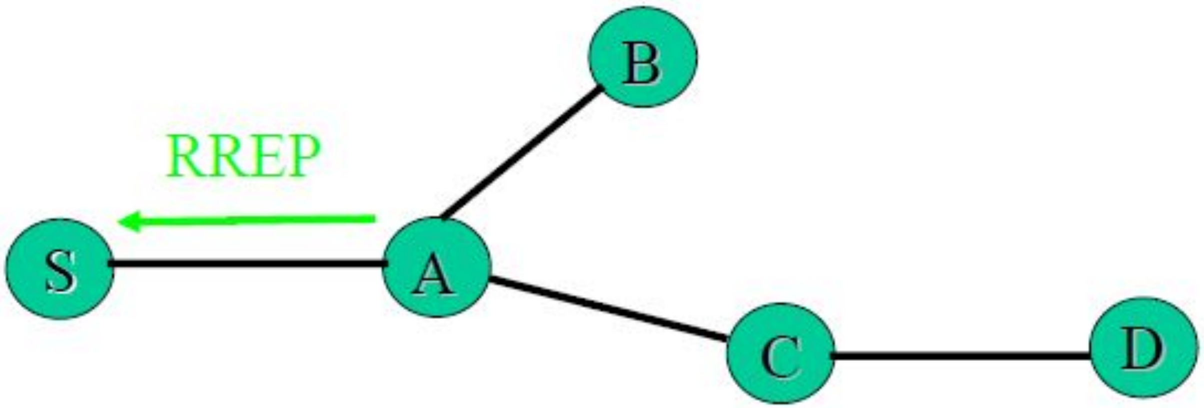
- When a node determines that it has a current route to respond to RREQ i.e. has a path to destination - It creates RREP (Route Reply)
- RREP contains <IP address of source and destination>
  - If RREP is being sent by destination
    - The RREP will also contain the <current sqn # of destination, hop-count=0, life-time>
  - If RREP is sent by an intermediate node
    - RREP will contain its record of the <destination sequence number, hop-count=its distance to destination, its value of the life-time>

Forward Path Setup

- When an intermediate node receives the RREP, it sets up a forward path entry to the destination in its route table
  - Forward path entry contains

  <IP Address of destination, IP address of node from which the entry arrived, hop-count to destination, life-time>
  - To obtain its distance to destination i.e. hop-count, a node increments its distance by 1
  - If route is not used within the life time, its deleted
- After processing the RREP, the node forwards it towards the source

Forward Path Setup

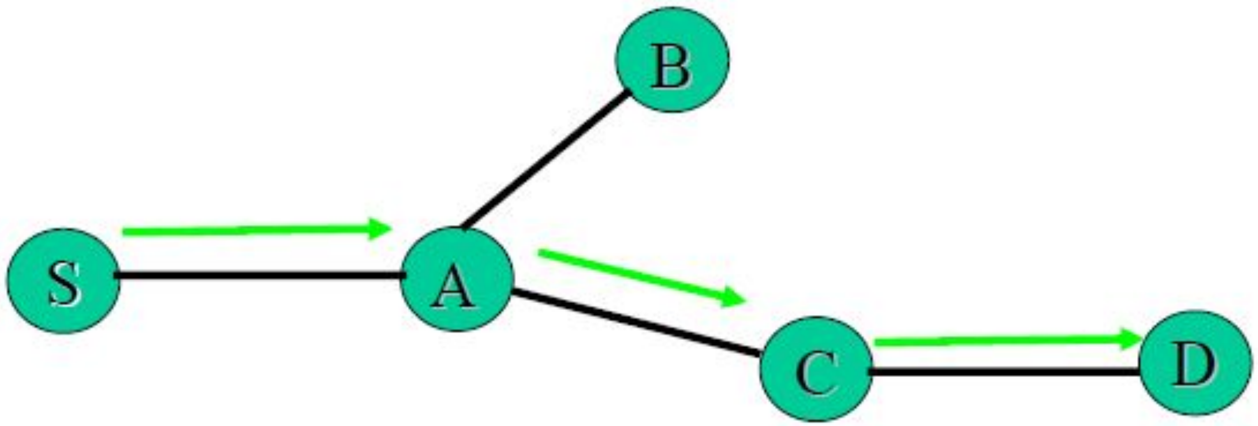


5. Node S receives RREP
   - Makes a forward route entry to D
     dest=D, nexthop =A, hopcount = 3

Receipt of Multiple RREP

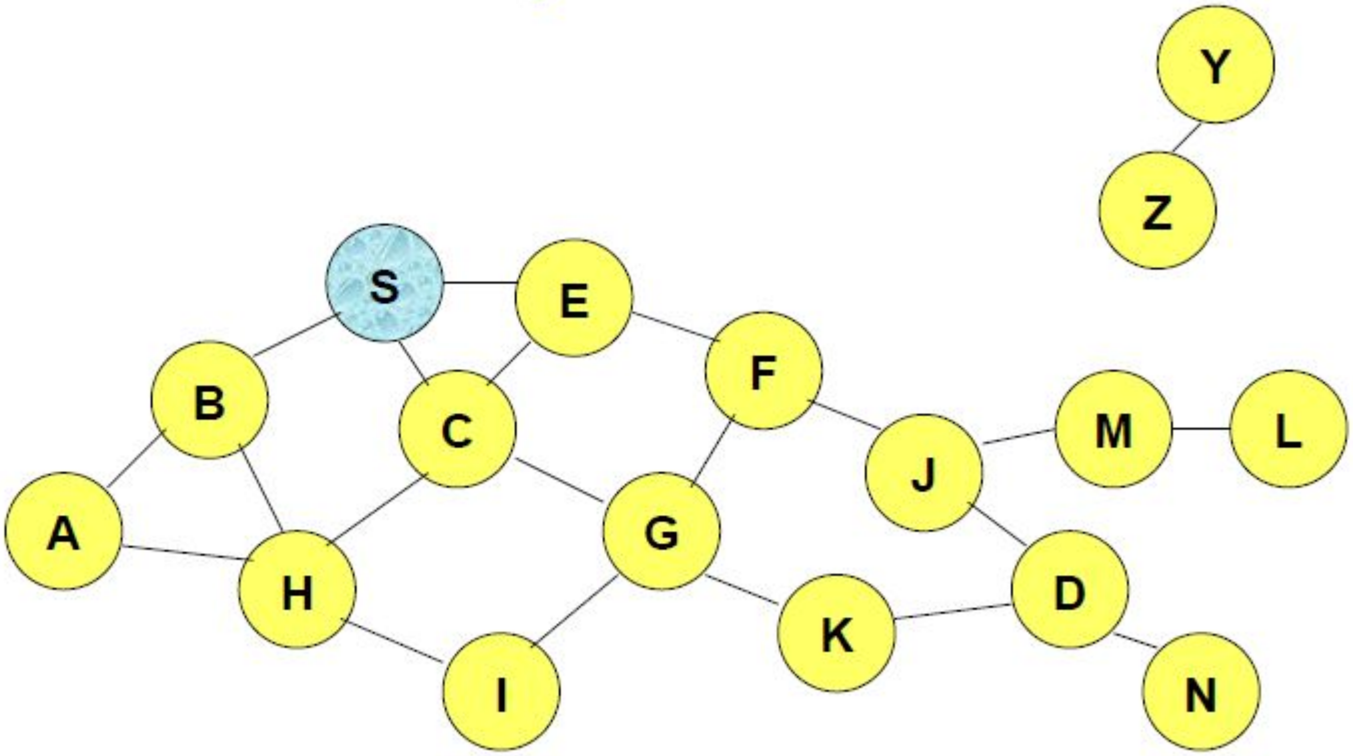
- A node may receive multiple RREP for a given destination from more than one neighbor
  - The node only forwards the first RREP it receives
  - May forward another RREP if that has greater destination sequence number or a smaller hop count
  - Rest are discarded -> reduces the number of RREP propagating towards the source
- Source can begin data transmission upon receiving the first RREP

AODV Data Delivery



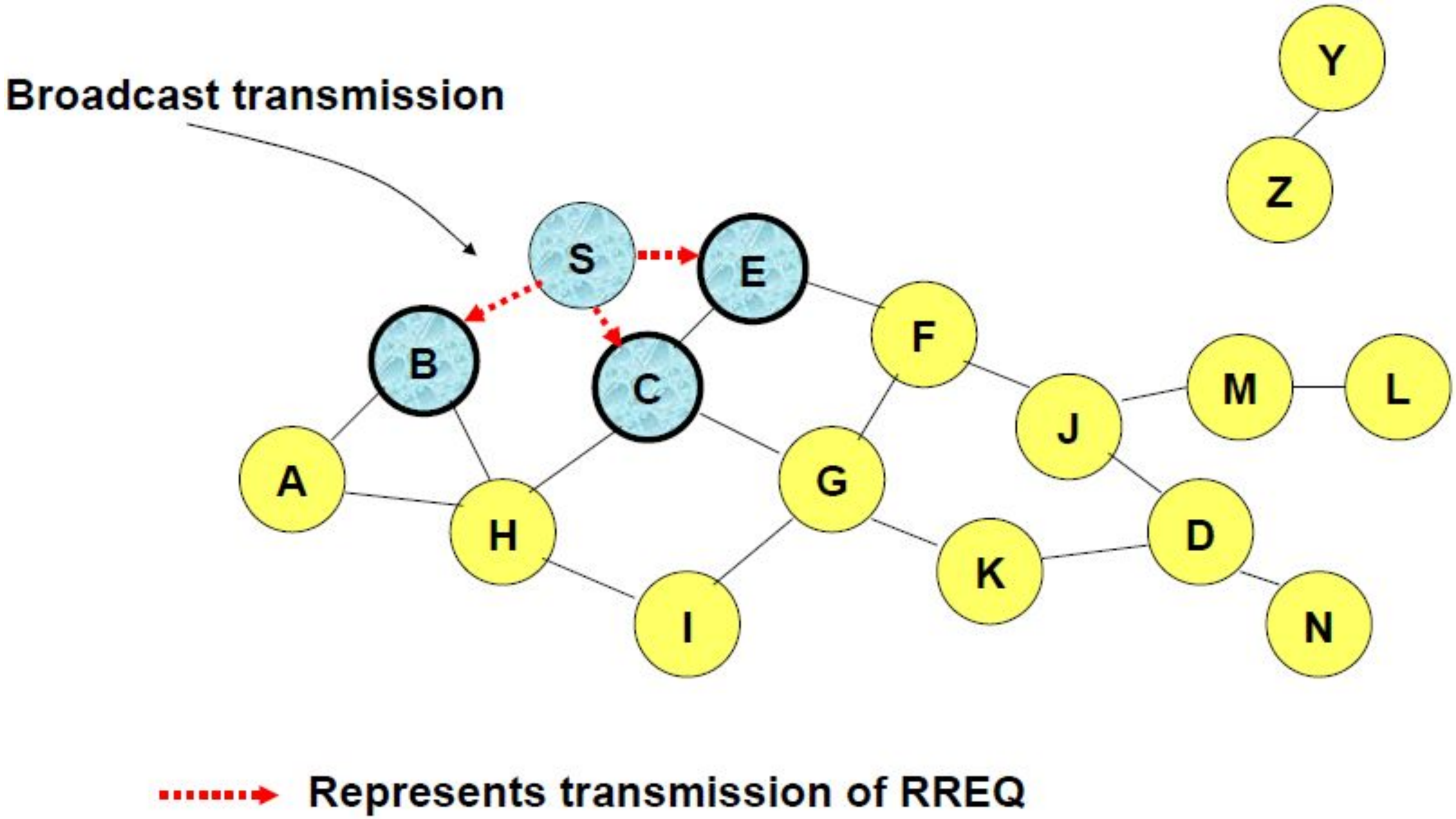
5. Node S receives RREP
   - Makes a forward route entry to D
     dest=D, nexthop =A, hopcount = 3
   - Sends data packet on route to D

Route Requests in AODV

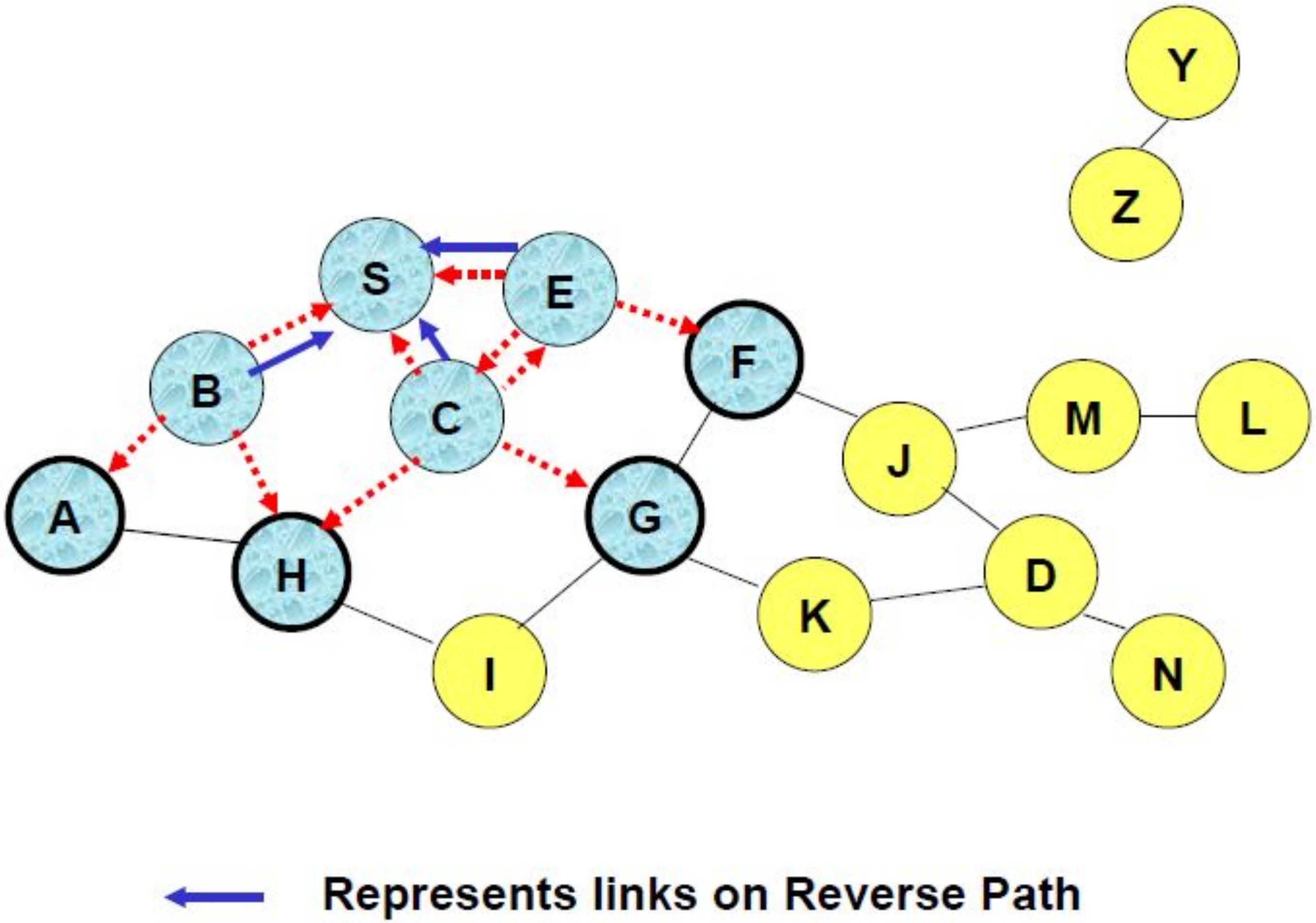


Represents a node that has received RREQ for D from S

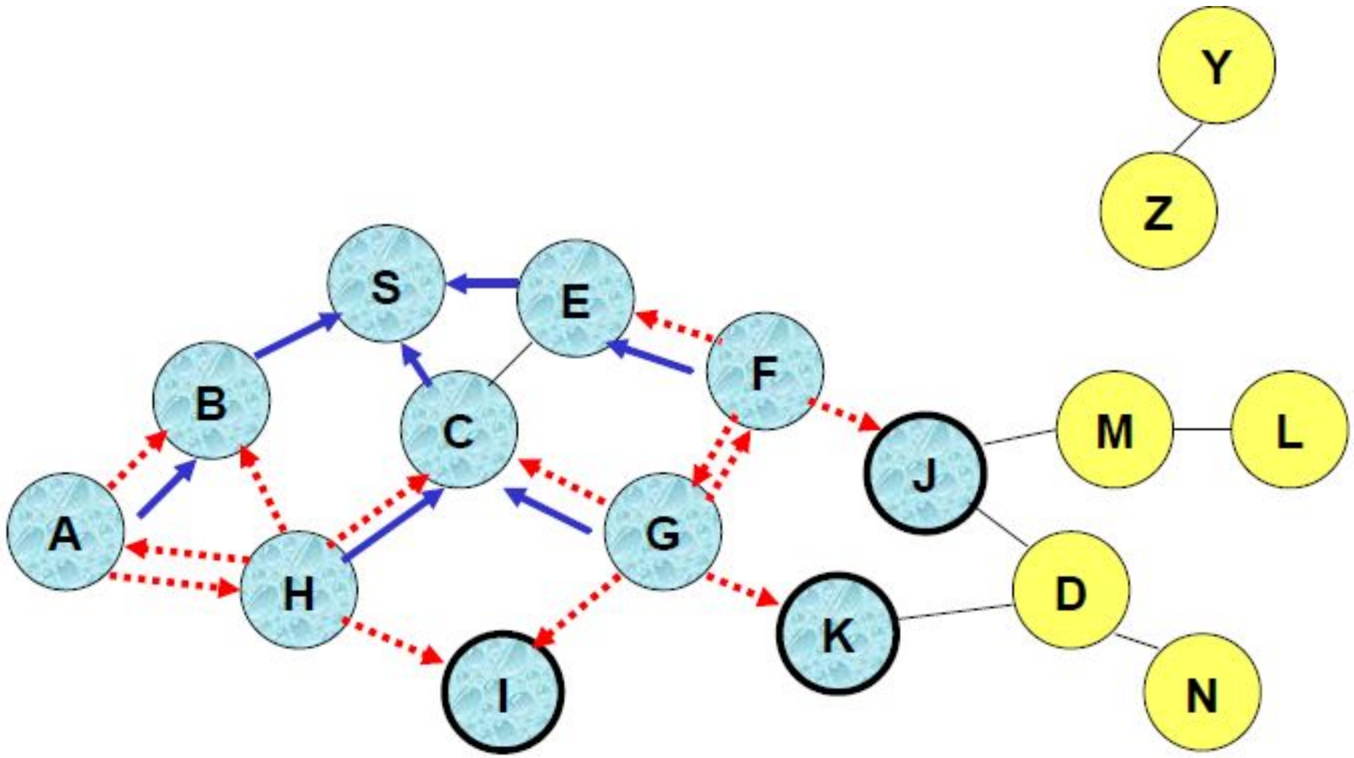
Route Requests in AODV



Route Requests in AODV

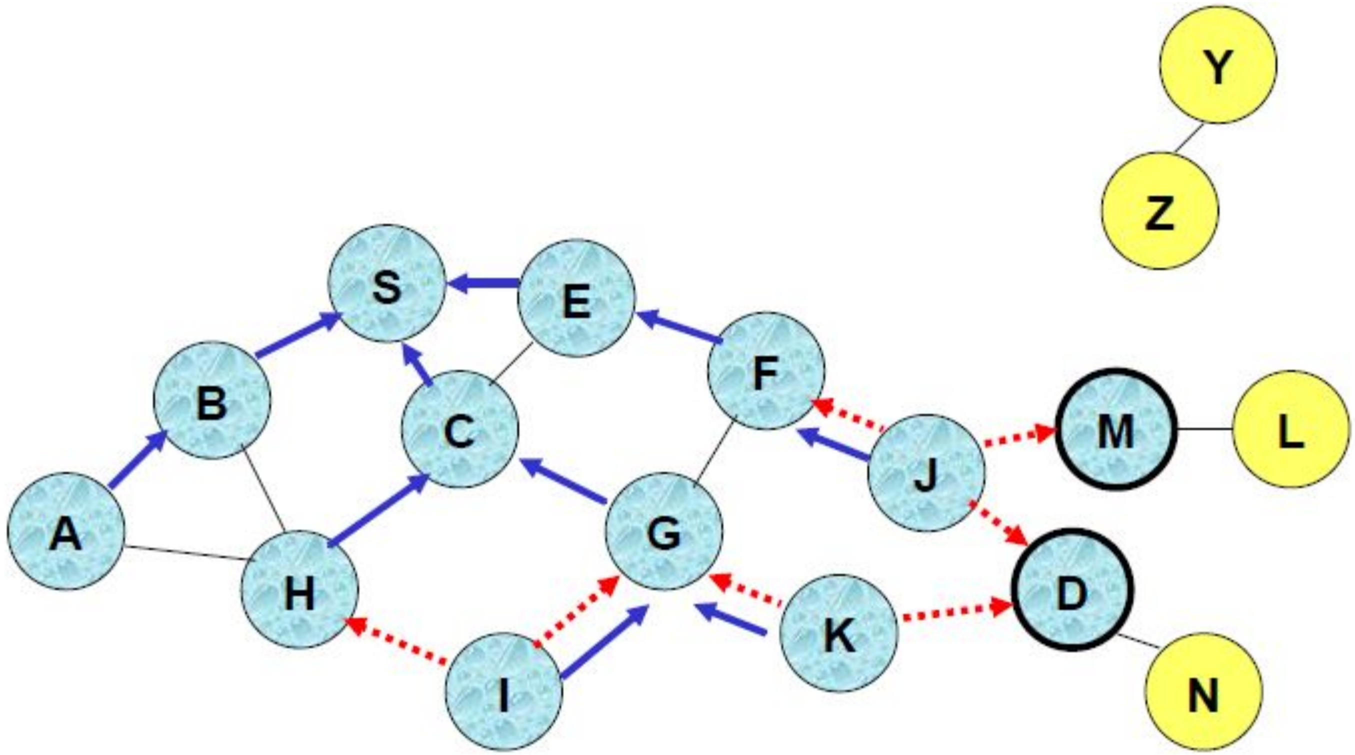


Reverse Path Setup in AODV

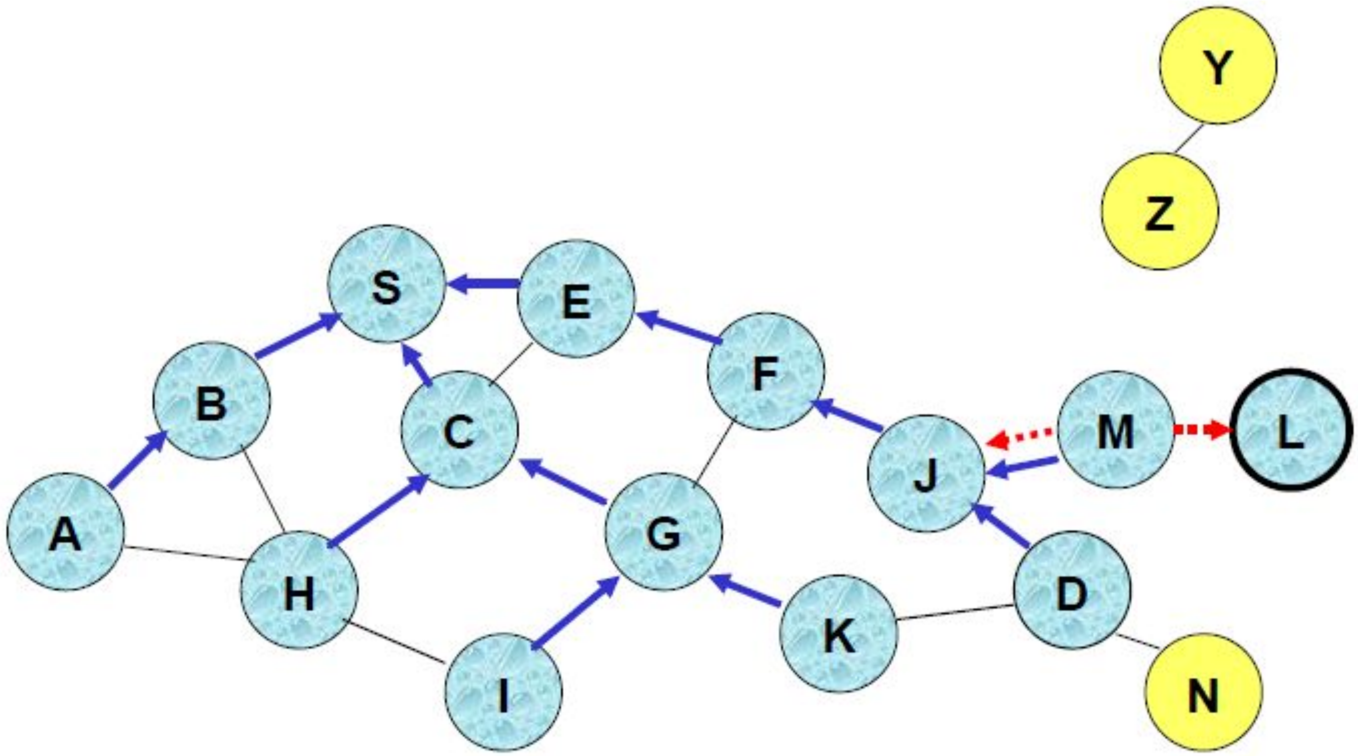


- Node C receives RREQ from G and H, but does not forward it again, because node C has already forwarded RREQ once

Reverse Path Setup in AODV

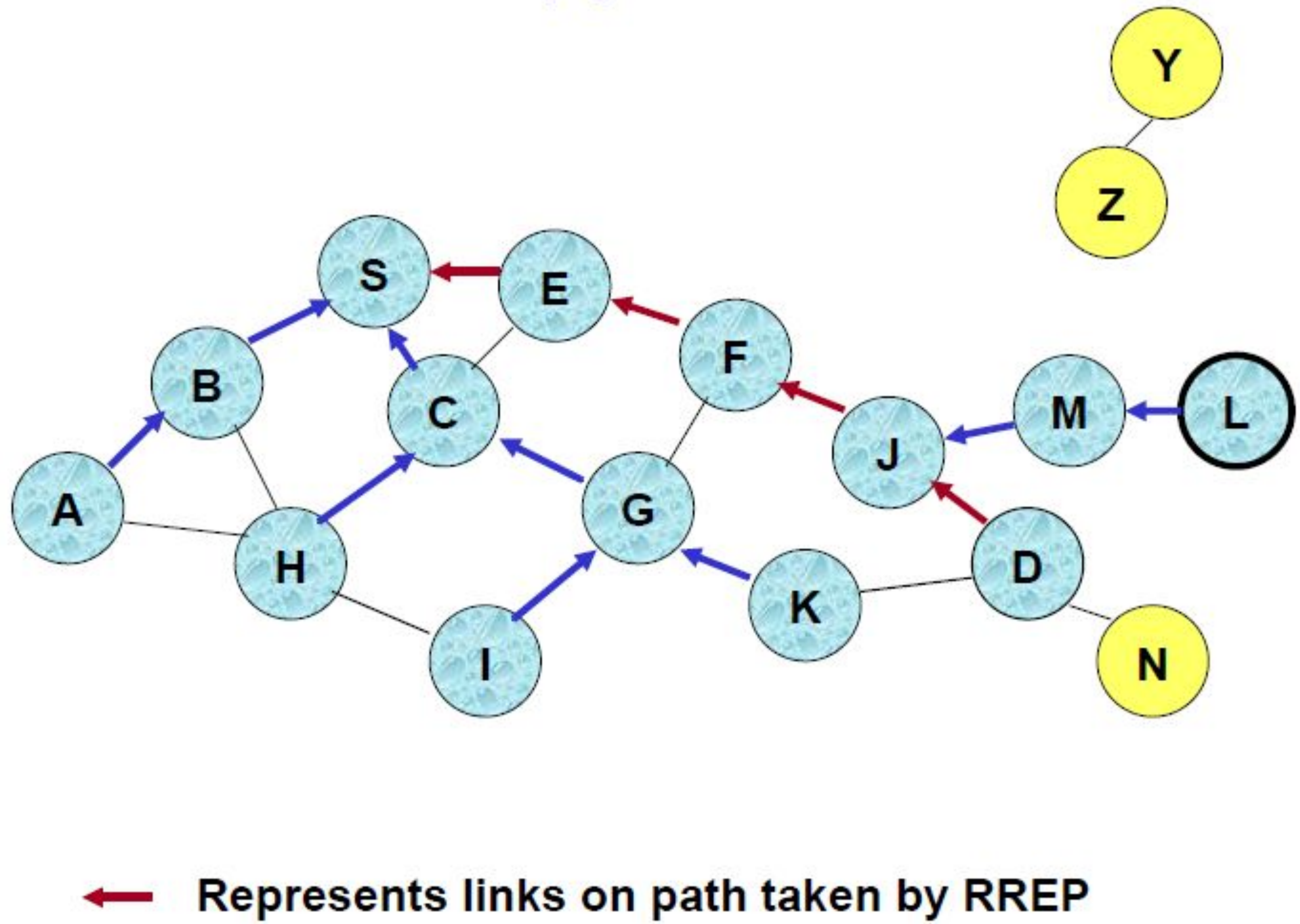


Reverse Path Setup in AODV



- Node D **does not forward** RREQ, because node D is the **intended target** of the RREQ

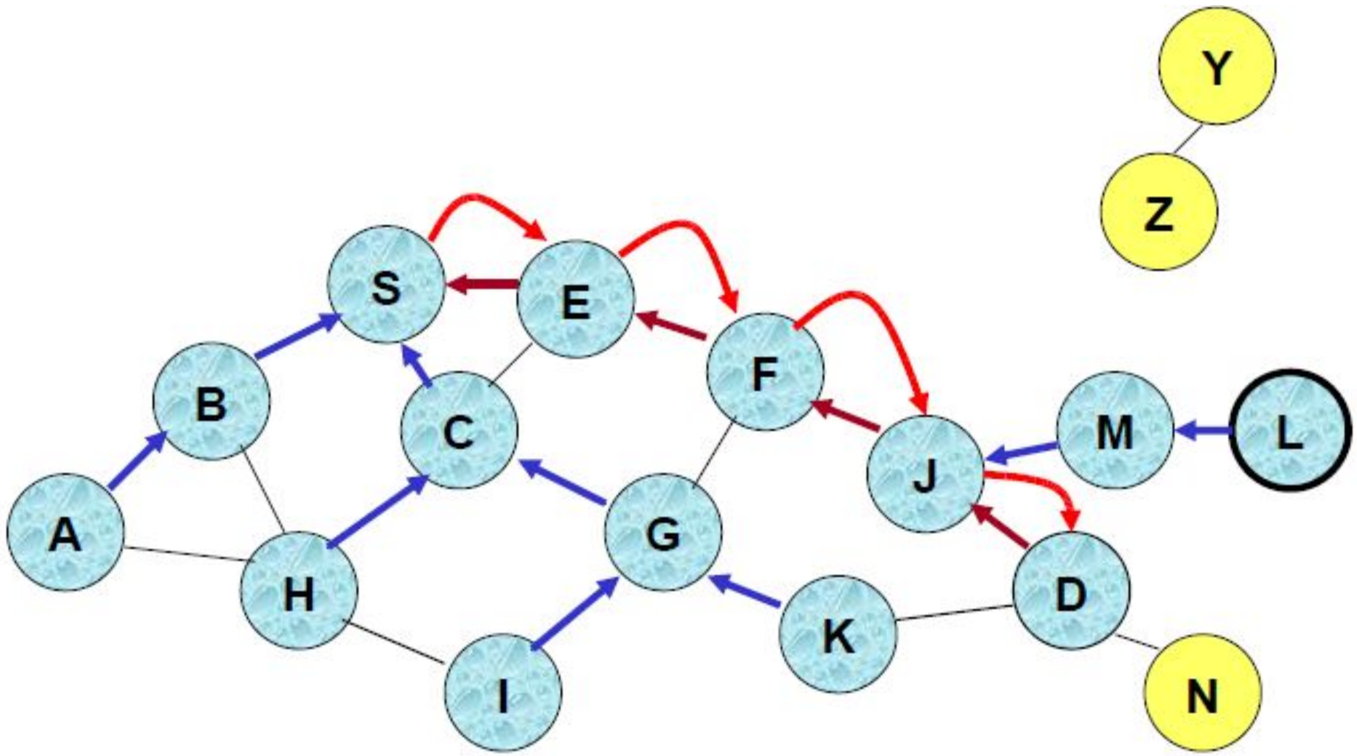
Route Reply in AODV



Route Reply in AODV

- An intermediate node (not the destination) may also send a Route Reply (RREP) provided that it knows a more recent path than the one previously known to sender S
- To determine whether the path known to an intermediate node is more recent, *destination sequence numbers are used*
- The likelihood that an intermediate node will send a Route Reply when using AODV not as high as DSR
  - A new Route Request by node S for a destination is assigned a higher destination sequence number. An intermediate node which knows a route, but with a smaller sequence number, cannot send Route Reply

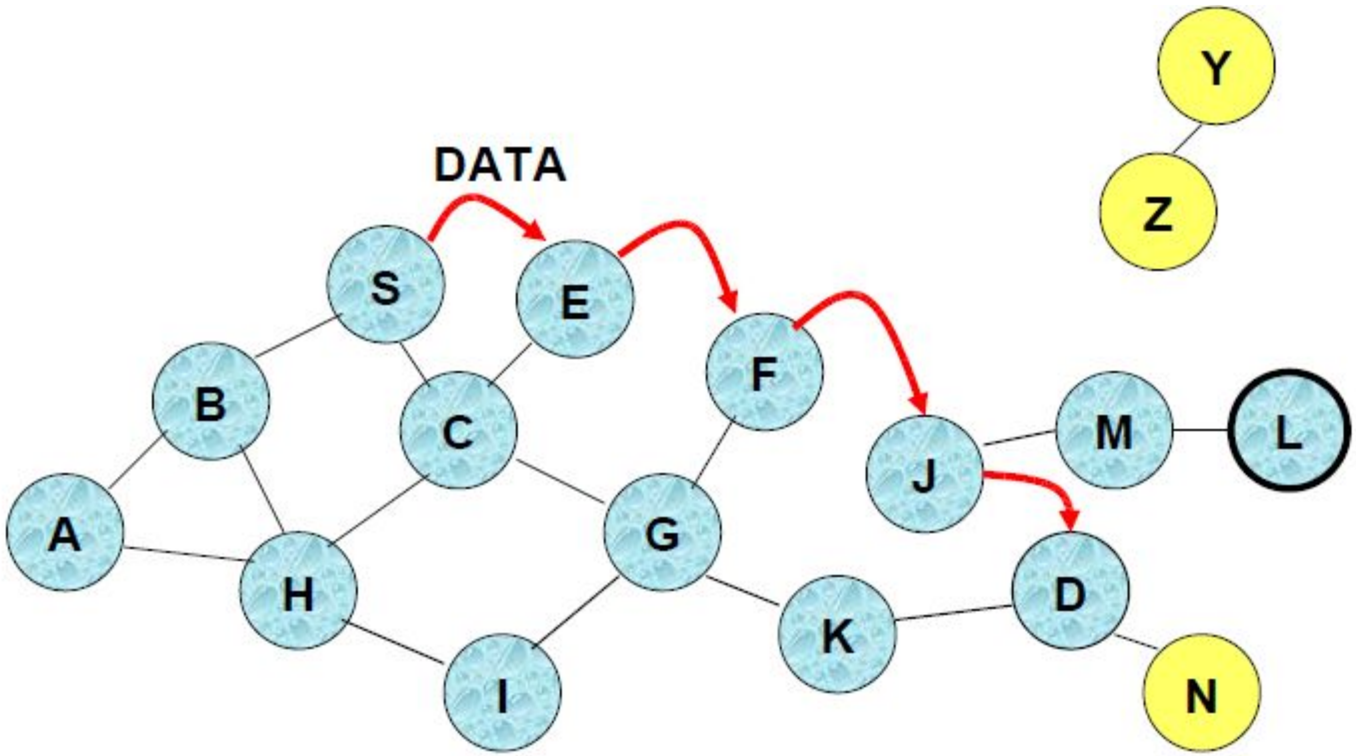
Forward Path Setup in AODV



Forward links are setup when RREP travels along the reverse path

Represents a link on the forward path

Data Delivery in AODV



Routing table entries used to forward data packet.

Route is *not* included in packet header.

Timeouts

- A routing table entry maintaining a reverse path is purged after a timeout interval
  - timeout should be long enough to allow RREP to come back
- A routing table entry maintaining a forward path is purged if *not used for active_route_timeout interval*
  - if no data is being sent using a particular routing table entry, that entry will be deleted from the routing table (even if the route may actually still be valid)

Link Failure Reporting

- A neighbor of node X is considered active for a routing table entry if the neighbor sent a packet within *active_route_timeout interval which was forwarded* using that entry
- When the next hop link in a routing table entry breaks, all active neighbors are informed
- Link failures are propagated by means of Route Error messages, which also update destination sequence numbers

Route Error

- When node X is unable to forward packet P (from node S to node D) on link (X,Y), it generates a RERR message
- Node X increments the destination sequence number for D cached at node X
- The incremented sequence number *N is included in* the RERR
- When node S receives the RERR, it initiates a new route discovery for D using destination sequence number at least as large as *N*

Destination Sequence Number

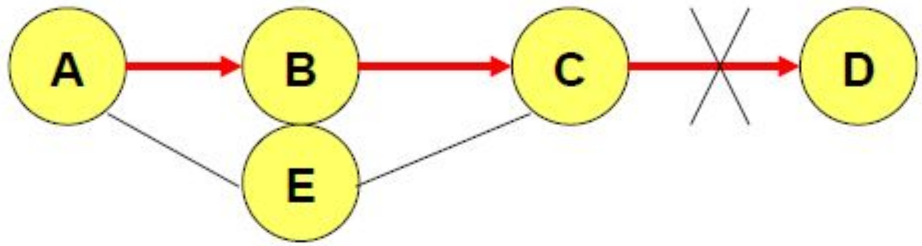
- Continuing from the previous slide …
  - When node D receives the route request with destination sequence number N, node D will set its sequence number to N, unless it is already larger than N

Link Failure Detection

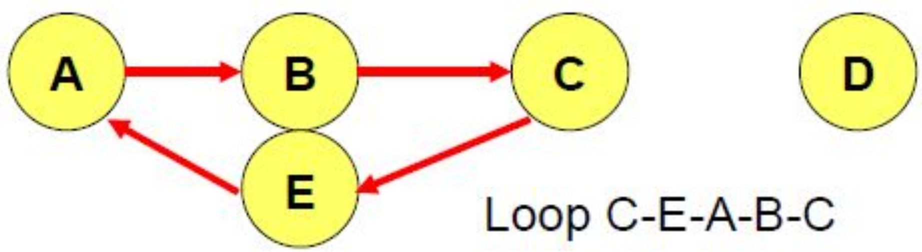
- *Hello messages: Neighboring nodes periodically* exchange hello message
- Absence of hello message is used as an indication of link failure
- Alternatively, failure to receive several MAC-level acknowledgement may be used as an indication of link failure

Why Sequence Numbers in AODV

- To avoid using old/broken routes
  - To determine which route is newer
- To prevent formation of loops



- Assume that A does not know about failure of link C-D because RERR sent by C is lost
- Now C performs a route discovery for D. Node A receives the RREQ (say, via path C-E-A)
- Node A will reply since A knows a route to D via node B Results in a loop (for instance, C-E-A-B-C)



Optimization: Expanding Ring Search

- Route Requests are initially sent with small Time-to-Live (TTL) field, to limit their propagation
  - DSR also includes a similar optimization
- If no Route Reply is received, then larger TTL tried

AODV: Summary

- Routes need not be included in packet headers
- Nodes maintain routing tables containing entries only for routes that are in active use
- At most one next-hop per destination maintained at each node
  - Multi-path extensions can be designed
  - DSR may maintain several routes for a single destination
- Unused routes expire even if topology does not change